

EDA Financial Discussions - Analysis Report

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Executive Summary

This report summarizes the exploratory data analysis conducted to identify suitable datasets for predicting engagement and sentiment surges in stock-related social media discussions.

Key Findings

- **Datasets discovered:** 0 (0 complete with engagement + sentiment fields)
- **API platforms assessed:** 2
- **Quality reports generated:** 11 (11 suitable)
- **Surge definitions evaluated:** 90 (48 viable with $\geq 2\%$ positive class)

Dataset Discovery Results

No datasets were discovered during the scan.

API Feasibility Findings

Twitter API

- **Historical access:** No
- **Supports surge label construction:** Yes
- **Estimated collection time:** 0.4 hours
- **Estimated cost:** \$100.00
- **Endpoints available:** 5

Rate Limits

- free_tier: {'tweets_per_month': 1500, 'requests_per_15min': 15, 'posts_per_request': 100}
- basic_tier: {'tweets_per_month': 10000, 'requests_per_15min': 60, 'posts_per_request': 100}
- pro_tier: {'tweets_per_month': 1000000, 'requests_per_15min': 300, 'posts_per_request': 100}

Cost Tiers

Tier	Details
Free	cost_usd_monthly: 0, tweet_cap: 1500
Basic	cost_usd_monthly: 100, tweet_cap: 10000
Pro	cost_usd_monthly: 5000, tweet_cap: 1000000
Enterprise	cost_usd_monthly: 42000, tweet_cap: 50000000

Paid Fields

The following fields require paid access:

- **impression_count**: requires Basic (\$100/month)
- **full archive search**: requires Pro (\$5,000/month)
- **quote_count**: requires Basic (\$100/month)

Reddit API

- **Historical access**: Yes
- **Supports surge label construction**: Yes
- **Estimated collection time**: 0.0 hours
- **Estimated cost**: \$0.00
- **Endpoints available**: 7

Rate Limits

- `oauth_tier`: {'requests_per_minute': 100, 'posts_per_request': 100, 'daily_limit': None}
- `free_tier_note`: Reddit API is free for non-commercial use with OAuth. Commercial use requires paid access.

Cost Tiers

Tier	Details
Free (non-commercial)	<code>cost_usd_monthly</code> : 0, <code>rate_limit</code> : 100 requests/min, <code>note</code> : Requires OAuth app registration
Commercial	<code>cost_usd_monthly</code> : Contact Reddit, <code>rate_limit</code> : Higher limits available, <code>note</code> : Required for commercial data use since 2023 API changes

Paid Fields

The following fields require paid access:

- **full historical archive**: requires Commercial (contact Reddit)
- **real-time streaming**: requires Commercial (contact Reddit)

EDA Statistics

`./data/StockMarket_subreddit.csv`

Dataset Structure

- **Records**: 72,620
- **Tickers**: 1597
- **Columns**: 9
- **Date range**: 1970-01-01 to 1970-01-01
- **Recommendation**: suitable

Missing Values

Column	Missing %
selftext	29.2%

Engagement Statistics

Metric	Mean	Median	P90	P95	P99
score	6.3	1.0	7.0	16.0	97.0
num_comments	8.1	1.0	17.0	32.0	124.0

Sentiment Statistics

- **mean:** 0.092
- **median:** 0.000
- **std:** 0.305
- **Bullish/Bearish ratio:** 2.48

Identified Risks

- Duplicate rows detected: 1103 duplicates (1.5%), which may inflate engagement statistics.

./data/goyaladi_twitter-dataset/goyaladi_twitter-dataset.csv

Dataset Structure

- **Records:** 10,000
- **Tickers:** 0
- **Columns:** 6
- **Date range:** 2023-01-01 to 2023-05-15
- **Recommendation:** suitable

Missing Values

Column	Missing %
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Engagement Statistics

Metric	Mean	Median	P90	P95	P99
Retweets	49.7	49.0	90.0	95.0	99.0
Likes	49.9	50.0	90.0	95.0	100.0

Sentiment Statistics

- **mean:** 0.443
- **median:** 0.599

- **std:** 0.452
- **Bullish/Bearish ratio:** 4.72

Identified Risks

- No ticker column 'ticker' found and ticker extraction from text yielded no results: per-ticker engagement normalization cannot be performed.

./data/leukipp_reddit-finance-data/gme/submissions_reddit.csv

Dataset Structure

- **Records:** 273,327
- **Tickers:** 340
- **Columns:** 25
- **Date range:** 2021-01-01 to 2021-12-31
- **Recommendation:** suitable

Missing Values

High-risk columns (>30% missing): selftext

Column	Missing %
selftext	48.7% 
link_flair_text	7.0%

Engagement Statistics

Metric	Mean	Median	P90	P95	P99
score	101.3	10.0	142.0	407.0	1868.7
num_comments	12.5	3.0	21.0	37.0	146.0

Sentiment Statistics

- **mean:** 0.083
- **median:** 0.000
- **std:** 0.361
- **Bullish/Bearish ratio:** 1.82

Identified Risks

- High missing data: columns ['selftext'] have >30% missing values, which may bias analysis or require imputation.

./data/leukipp_reddit-finance-data/investing/submissions_reddit.csv

Dataset Structure

- **Records:** 41,912
- **Tickers:** 990
- **Columns:** 25
- **Date range:** 2021-01-01 to 2021-12-31
- **Recommendation:** suitable

Missing Values

High-risk columns (>30% missing): link_flair_text

Column	Missing %
link_flair_text	100.0% ⚠
selftext	0.0%

Engagement Statistics

Metric	Mean	Median	P90	P95	P99
score	17.8	1.0	7.0	25.0	377.7
num_comments	10.8	1.0	16.0	40.0	201.9

Sentiment Statistics

- **mean:** 0.111
- **median:** 0.000
- **std:** 0.304
- **Bullish/Bearish ratio:** 2.84

Identified Risks

- High missing data: columns ['link_flair_text'] have >30% missing values, which may bias analysis or require imputation.

./data/leukipp_reddit-finance-data/options/submissions_reddit.csv

Dataset Structure

- **Records:** 28,782
- **Tickers:** 624
- **Columns:** 25
- **Date range:** 2021-01-01 to 2021-12-31
- **Recommendation:** suitable

Missing Values

High-risk columns (>30% missing): link_flair_text

Column	Missing %
link_flair_text	100.0% 
selftext	11.6%

Engagement Statistics

Metric	Mean	Median	P90	P95	P99
score	11.3	1.0	9.0	24.0	208.0
num_comments	10.4	3.0	21.0	36.0	128.0

Sentiment Statistics

- **mean:** 0.074
- **median:** 0.000
- **std:** 0.286
- **Bullish/Bearish ratio:** 2.03

Identified Risks

- High missing data: columns ['link_flair_text'] have >30% missing values, which may bias analysis or require imputation.

./data/leukipp_reddit-finance-data/pennystocks/submissions_reddit.csv

Dataset Structure

- **Records:** 54,785
- **Tickers:** 2912
- **Columns:** 25
- **Date range:** 2021-01-01 to 2021-12-31
- **Recommendation:** suitable

Missing Values

Column	Missing %
selftext	20.7%
link_flair_text	1.0%

Engagement Statistics

Metric	Mean	Median	P90	P95	P99
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Metric	Mean	Median	P90	P95	P99
score	29.4	1.0	21.0	56.8	437.5
num_comments	10.8	1.0	21.0	39.0	137.0

Sentiment Statistics

- **mean:** 0.118
- **median:** 0.000
- **std:** 0.298
- **Bullish/Bearish ratio:** 3.50

./data/leukipp_reddit-finance-data/robinhood/submissions_reddit.csv

Dataset Structure

- **Records:** 18,893
- **Tickers:** 294
- **Columns:** 25
- **Date range:** 2021-01-01 to 2021-12-31
- **Recommendation:** suitable

Missing Values

High-risk columns (>30% missing): link_flair_text

Column	Missing %
link_flair_text	47.8% ⚠
selftext	25.3%

Engagement Statistics

Metric	Mean	Median	P90	P95	P99
score	5.5	1.0	1.0	1.0	54.1
num_comments	2.2	0.0	1.0	3.0	37.0

Sentiment Statistics

- **mean:** 0.049
- **median:** 0.000
- **std:** 0.340
- **Bullish/Bearish ratio:** 1.42

Identified Risks

- High missing data: columns ['link_flair_text'] have >30% missing values, which may bias analysis or require imputation.

./data/leukipp_reddit-finance-data/robinhoodpennystocks/submissions_reddit.csv

Dataset Structure

- **Records:** 23,304
- **Tickers:** 855
- **Columns:** 25
- **Date range:** 2021-01-01 to 2021-12-31
- **Recommendation:** suitable

Missing Values

High-risk columns (>30% missing): link_flair_text, selftext

Column	Missing %
link_flair_text	46.8% 
selftext	36.0% 

Engagement Statistics

Metric	Mean	Median	P90	P95	P99
score	30.8	1.0	28.7	70.0	507.9
num_comments	9.1	0.0	22.0	39.0	115.0

Sentiment Statistics

- **mean:** 0.098
- **median:** 0.000
- **std:** 0.312
- **Bullish/Bearish ratio:** 2.61

Identified Risks

- High missing data: columns ['link_flair_text', 'selftext'] have >30% missing values, which may bias analysis or require imputation.

./data/leukipp_reddit-finance-data/stockmarket/submissions_reddit.csv

Dataset Structure

- **Records:** 43,809
- **Tickers:** 1475
- **Columns:** 25

- **Date range:** 2021-01-01 to 2021-12-31
- **Recommendation:** suitable

Missing Values

High-risk columns (>30% missing): selftext

Column	Missing %
selftext	39.1% 
link_flair_text	13.8%

Engagement Statistics

Metric	Mean	Median	P90	P95	P99
score	35.0	1.0	18.0	65.0	848.9
num_comments	7.0	0.0	13.0	28.0	129.0

Sentiment Statistics

- **mean:** 0.104
- **median:** 0.000
- **std:** 0.318
- **Bullish/Bearish ratio:** 2.59

Identified Risks

- High missing data: columns ['selftext'] have >30% missing values, which may bias analysis or require imputation.

./data/leukipp_reddit-finance-data/stocks/submissions_reddit.csv

Dataset Structure

- **Records:** 75,857
- **Tickers:** 1963
- **Columns:** 25
- **Date range:** 2021-01-01 to 2021-12-31
- **Recommendation:** suitable

Missing Values

High-risk columns (>30% missing): link_flair_text

Column	Missing %
link_flair_text	46.9% 

Column	Missing %
selftext	0.1%

Engagement Statistics

Metric	Mean	Median	P90	P95	P99
score	30.2	1.0	14.0	39.0	413.0
num_comments	14.8	1.0	26.0	51.0	250.4

Sentiment Statistics

- **mean:** 0.085
- **median:** 0.000
- **std:** 0.282
- **Bullish/Bearish ratio:** 2.53

Identified Risks

- High missing data: columns ['link_flair_text'] have >30% missing values, which may bias analysis or require imputation.

./data/leukipp_reddit-finance-data/wallstreetbets/submissions_reddit.csv

Dataset Structure

- **Records:** 775,326
- **Tickers:** 4451
- **Columns:** 25
- **Date range:** 2021-01-01 to 2021-12-31
- **Recommendation:** suitable

Missing Values

High-risk columns (> 30% missing): selftext

Column	Missing %
selftext	33.8% ⚠
link_flair_text	1.7%
title	0.0%

Engagement Statistics

Metric	Mean	Median	P90	P95	P99
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Metric	Mean	Median	P90	P95	P99
score	116.0	1.0	21.0	70.0	964.0
num_comments	24.6	1.0	11.0	29.0	164.0

Sentiment Statistics

- **mean:** 0.070
- **median:** 0.000
- **std:** 0.340
- **Bullish/Bearish ratio:** 1.80

Identified Risks

- High missing data: columns ['selftext'] have >30% missing values, which may bias analysis or require imputation.

Surge Analysis Results

Surge definitions were evaluated across multiple threshold combinations.

Engagement Percentile	Sentiment Std Devs	Window (hrs)	Surge Count	Total Posts	Surge %	Imbalance Ratio	Viable
0.90	0.5	24	6,171	72,620	8.50%	10.8:1	✓
0.90	1.0	24	4,411	72,620	6.07%	15.5:1	✓
0.90	1.5	24	2,483	72,620	3.42%	28.2:1	✓
0.95	0.5	24	3,521	72,620	4.85%	19.6:1	✓
0.95	1.0	24	2,388	72,620	3.29%	29.4:1	✓
0.95	1.5	24	1,371	72,620	1.89%	52.0:1	X
0.99	0.5	24	1,505	72,620	2.07%	47.3:1	✓
0.99	1.0	24	794	72,620	1.09%	90.5:1	X
0.99	1.5	24	418	72,620	0.58%	172.7:1	X
0.90	0.5	24	20,350	273,327	7.45%	12.4:1	✓
0.90	1.0	24	14,104	273,327	5.16%	18.4:1	✓
0.90	1.5	24	8,176	273,327	2.99%	32.4:1	✓
0.95	0.5	24	10,219	273,327	3.74%	25.7:1	✓
0.95	1.0	24	7,146	273,327	2.61%	37.2:1	✓
0.95	1.5	24	4,187	273,327	1.53%	64.3:1	X
0.99	0.5	24	2,126	273,327	0.78%	127.6:1	X

Engagement Percentile	Sentiment Std Devs	Window (hrs)	Surge Count	Total Posts	Surge %	Imbalance Ratio	Viable
0.99	1.0	24	1,558	273,327	0.57%	174.4:1	X
0.99	1.5	24	968	273,327	0.35%	281.4:1	X
0.90	0.5	24	2,255	41,912	5.38%	17.6:1	✓
0.90	1.0	24	1,341	41,912	3.20%	30.3:1	✓
0.90	1.5	24	730	41,912	1.74%	56.4:1	X
0.95	0.5	24	1,225	41,912	2.92%	33.2:1	✓
0.95	1.0	24	697	41,912	1.66%	59.1:1	X
0.95	1.5	24	381	41,912	0.91%	109.0:1	X
0.99	0.5	24	352	41,912	0.84%	118.1:1	X
0.99	1.0	24	190	41,912	0.45%	219.6:1	X
0.99	1.5	24	98	41,912	0.23%	426.7:1	X
0.90	0.5	24	1,429	28,782	4.96%	19.1:1	✓
0.90	1.0	24	1,078	28,782	3.75%	25.7:1	✓
0.90	1.5	24	588	28,782	2.04%	47.9:1	✓
0.95	0.5	24	728	28,782	2.53%	38.5:1	✓
0.95	1.0	24	548	28,782	1.90%	51.5:1	X
0.95	1.5	24	330	28,782	1.15%	86.2:1	X
0.99	0.5	24	183	28,782	0.64%	156.3:1	X
0.99	1.0	24	132	28,782	0.46%	217.0:1	X
0.99	1.5	24	80	28,782	0.28%	358.8:1	X
0.90	0.5	24	2,869	54,785	5.24%	18.1:1	✓
0.90	1.0	24	2,006	54,785	3.66%	26.3:1	✓
0.90	1.5	24	1,092	54,785	1.99%	49.2:1	X
0.95	0.5	24	1,634	54,785	2.98%	32.5:1	✓
0.95	1.0	24	1,106	54,785	2.02%	48.5:1	✓
0.95	1.5	24	589	54,785	1.08%	92.0:1	X
0.99	0.5	24	592	54,785	1.08%	91.5:1	X
0.99	1.0	24	356	54,785	0.65%	152.9:1	X
0.99	1.5	24	163	54,785	0.30%	335.1:1	X

Engagement Percentile	Sentiment Std Devs	Window (hrs)	Surge Count	Total Posts	Surge %	Imbalance Ratio	Viable
0.90	0.5	24	8,172	18,893	43.25%	1.3:1	✓
0.90	1.0	24	6,105	18,893	32.31%	2.1:1	✓
0.90	1.5	24	3,031	18,893	16.04%	5.2:1	✓
0.95	0.5	24	8,112	18,893	42.94%	1.3:1	✓
0.95	1.0	24	6,062	18,893	32.09%	2.1:1	✓
0.95	1.5	24	3,016	18,893	15.96%	5.3:1	✓
0.99	0.5	24	106	18,893	0.56%	177.2:1	✗
0.99	1.0	24	70	18,893	0.37%	268.9:1	✗
0.99	1.5	24	30	18,893	0.16%	628.8:1	✗
0.90	0.5	24	1,352	23,304	5.80%	16.2:1	✓
0.90	1.0	24	994	23,304	4.27%	22.4:1	✓
0.90	1.5	24	613	23,304	2.63%	37.0:1	✓
0.95	0.5	24	731	23,304	3.14%	30.9:1	✓
0.95	1.0	24	506	23,304	2.17%	45.1:1	✓
0.95	1.5	24	304	23,304	1.30%	75.7:1	✗
0.99	0.5	24	220	23,304	0.94%	104.9:1	✗
0.99	1.0	24	117	23,304	0.50%	198.2:1	✗
0.99	1.5	24	64	23,304	0.27%	363.1:1	✗
0.90	0.5	24	2,645	43,809	6.04%	15.6:1	✓
0.90	1.0	24	1,948	43,809	4.45%	21.5:1	✓
0.90	1.5	24	1,101	43,809	2.51%	38.8:1	✓
0.95	0.5	24	1,344	43,809	3.07%	31.6:1	✓
0.95	1.0	24	957	43,809	2.18%	44.8:1	✓
0.95	1.5	24	544	43,809	1.24%	79.5:1	✗
0.99	0.5	24	366	43,809	0.84%	118.7:1	✗
0.99	1.0	24	211	43,809	0.48%	206.6:1	✗
0.99	1.5	24	109	43,809	0.25%	400.9:1	✗
0.90	0.5	24	4,088	75,857	5.39%	17.6:1	✓
0.90	1.0	24	3,176	75,857	4.19%	22.9:1	✓

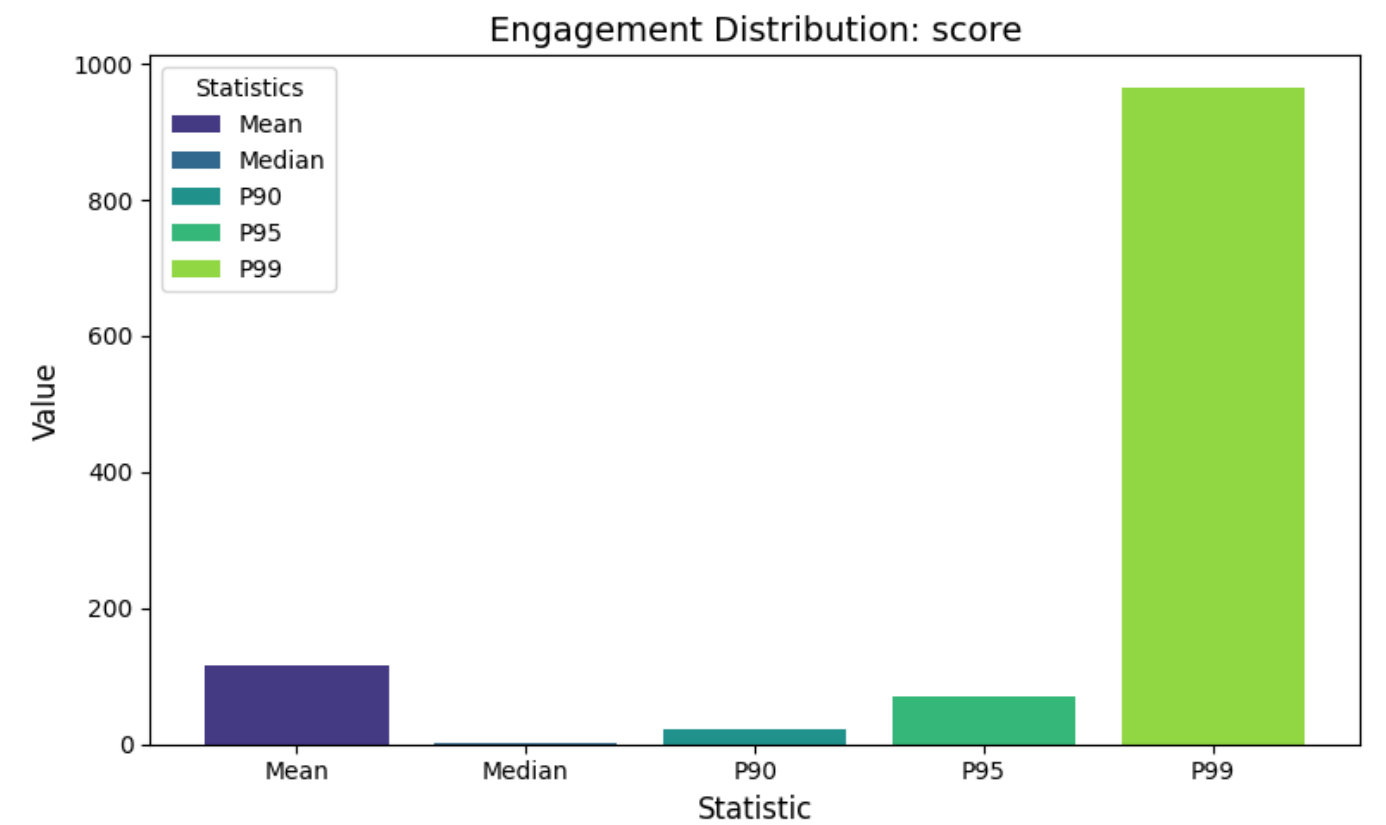
Engagement Percentile	Sentiment Std Devs	Window (hrs)	Surge Count	Total Posts	Surge %	Imbalance Ratio	Viable
0.90	1.5	24	1,864	75,857	2.46%	39.7:1	✓
0.95	0.5	24	2,126	75,857	2.80%	34.7:1	✓
0.95	1.0	24	1,626	75,857	2.14%	45.7:1	✓
0.95	1.5	24	952	75,857	1.25%	78.7:1	X
0.99	0.5	24	569	75,857	0.75%	132.3:1	X
0.99	1.0	24	379	75,857	0.50%	199.2:1	X
0.99	1.5	24	207	75,857	0.27%	365.5:1	X
0.90	0.5	24	52,040	775,326	6.71%	13.9:1	✓
0.90	1.0	24	37,925	775,326	4.89%	19.4:1	✓
0.90	1.5	24	21,228	775,326	2.74%	35.5:1	✓
0.95	0.5	24	26,971	775,326	3.48%	27.7:1	✓
0.95	1.0	24	19,574	775,326	2.52%	38.6:1	✓
0.95	1.5	24	10,901	775,326	1.41%	70.1:1	X
0.99	0.5	24	6,010	775,326	0.78%	128.0:1	X
0.99	1.0	24	4,048	775,326	0.52%	190.5:1	X
0.99	1.5	24	2,186	775,326	0.28%	353.7:1	X

48 viable surge definition(s) found (positive class $\geq$ 2%).

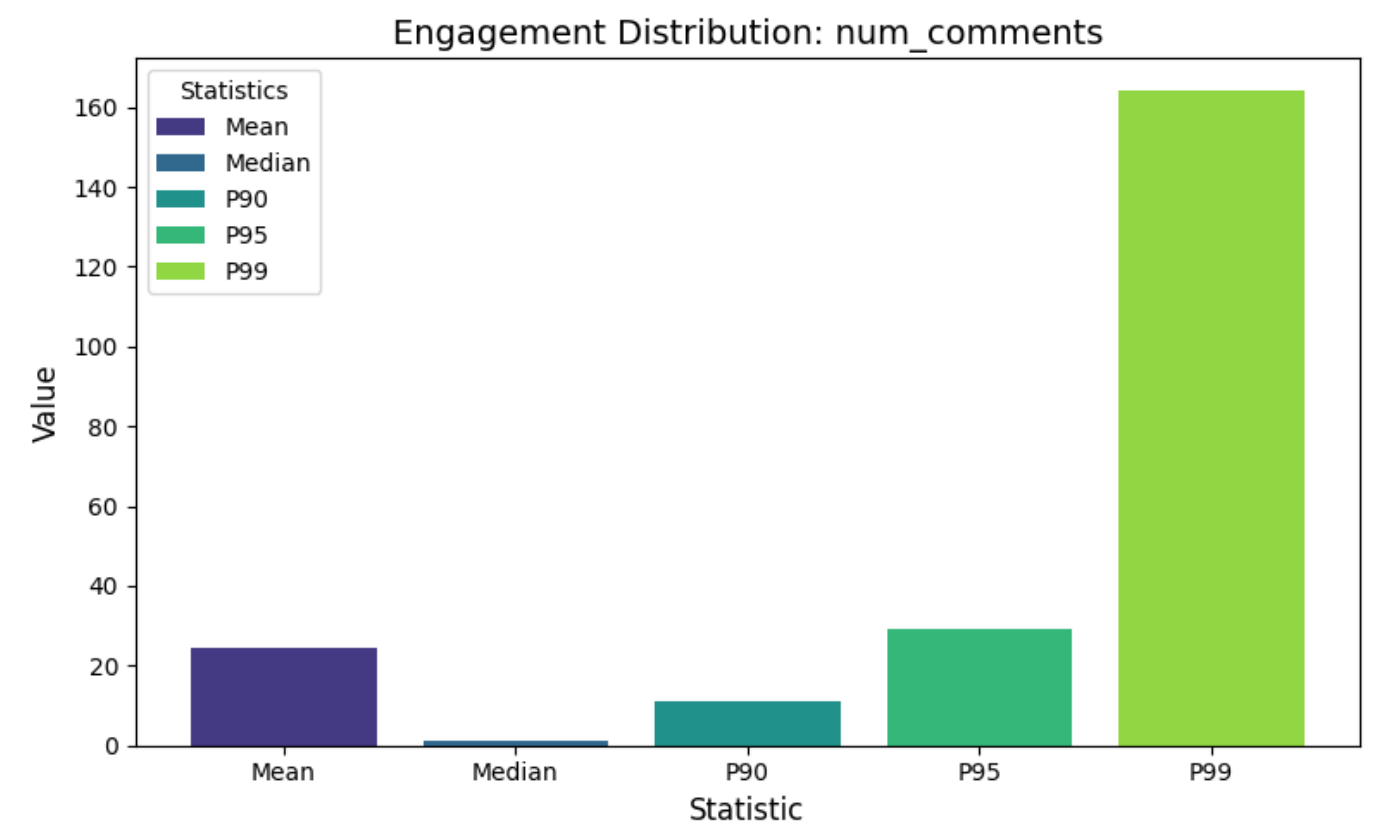
Best viable definition: engagement $\geq$ 90% percentile + sentiment shift $\geq$ 0.5 std devs $\rightarrow$ 43.25% surge rate.

Visualizations

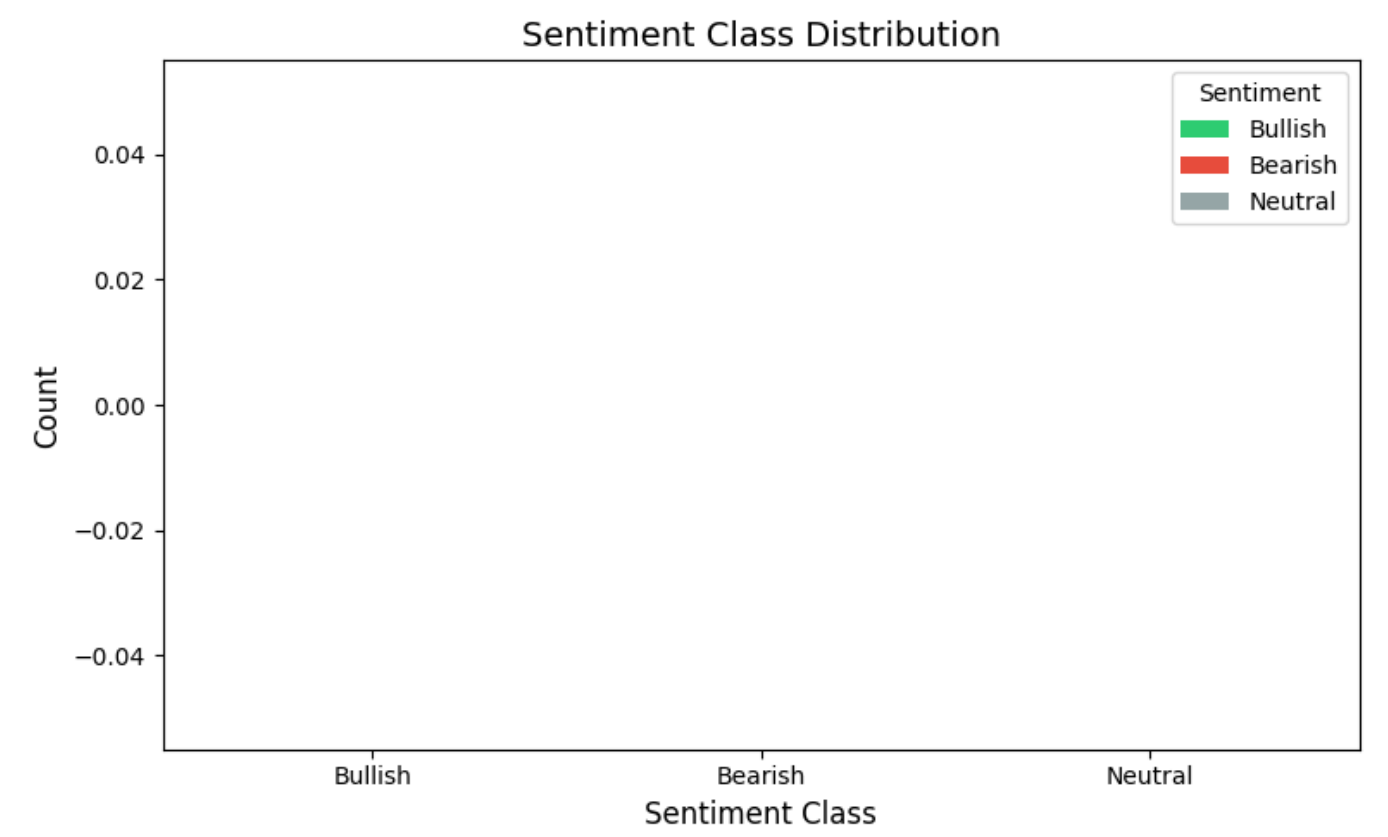
Engagement Distribution Score



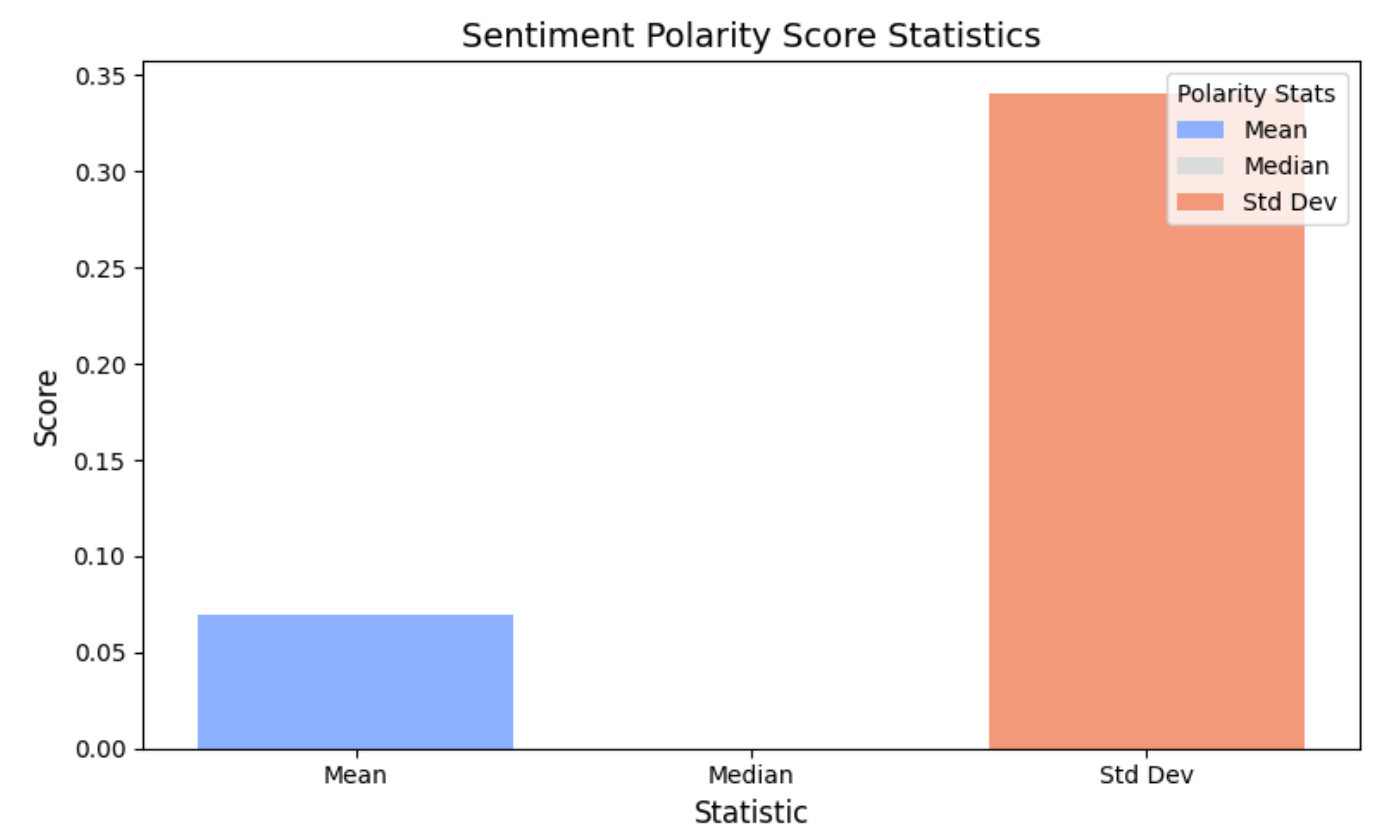
Engagement Distribution Num Comments



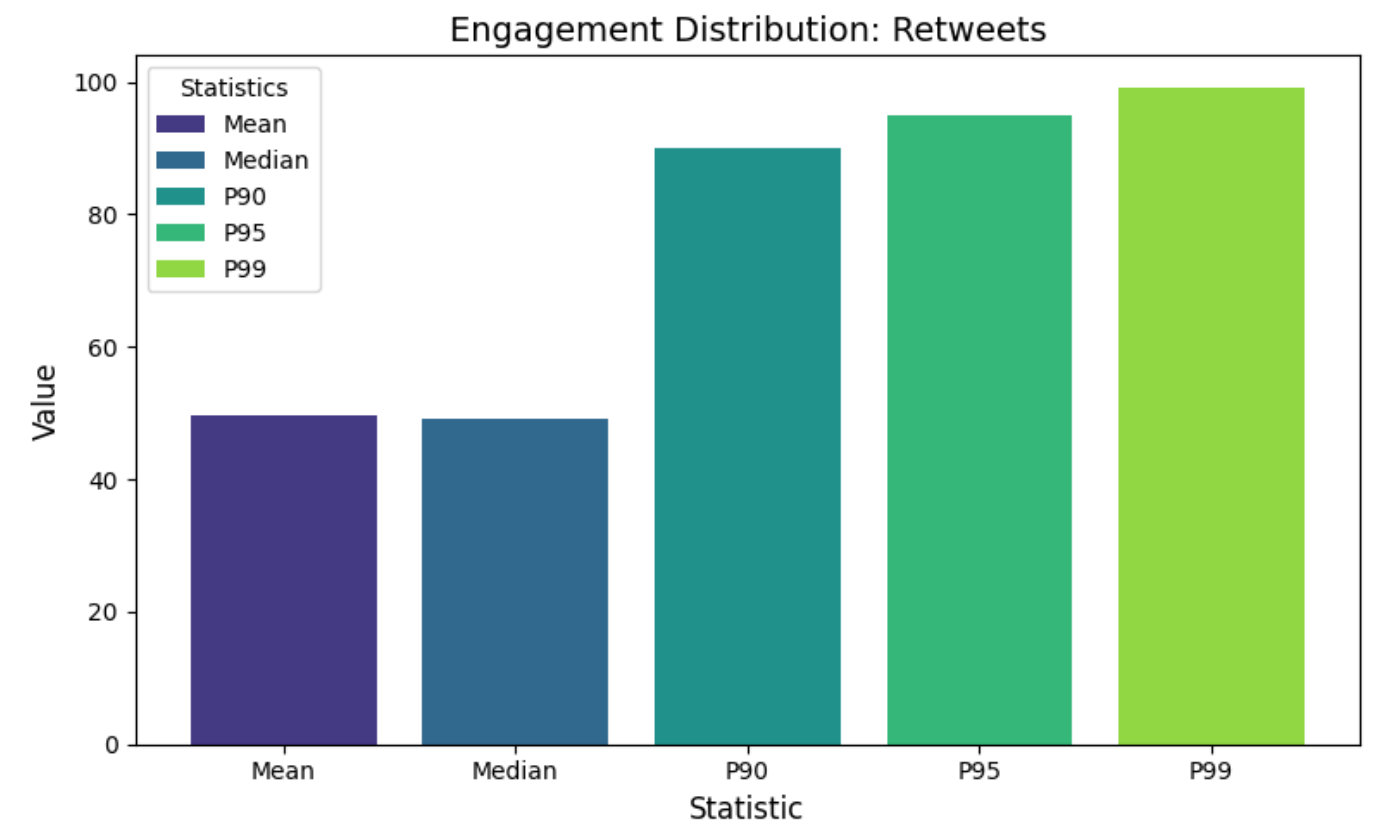
Sentiment Class Distribution



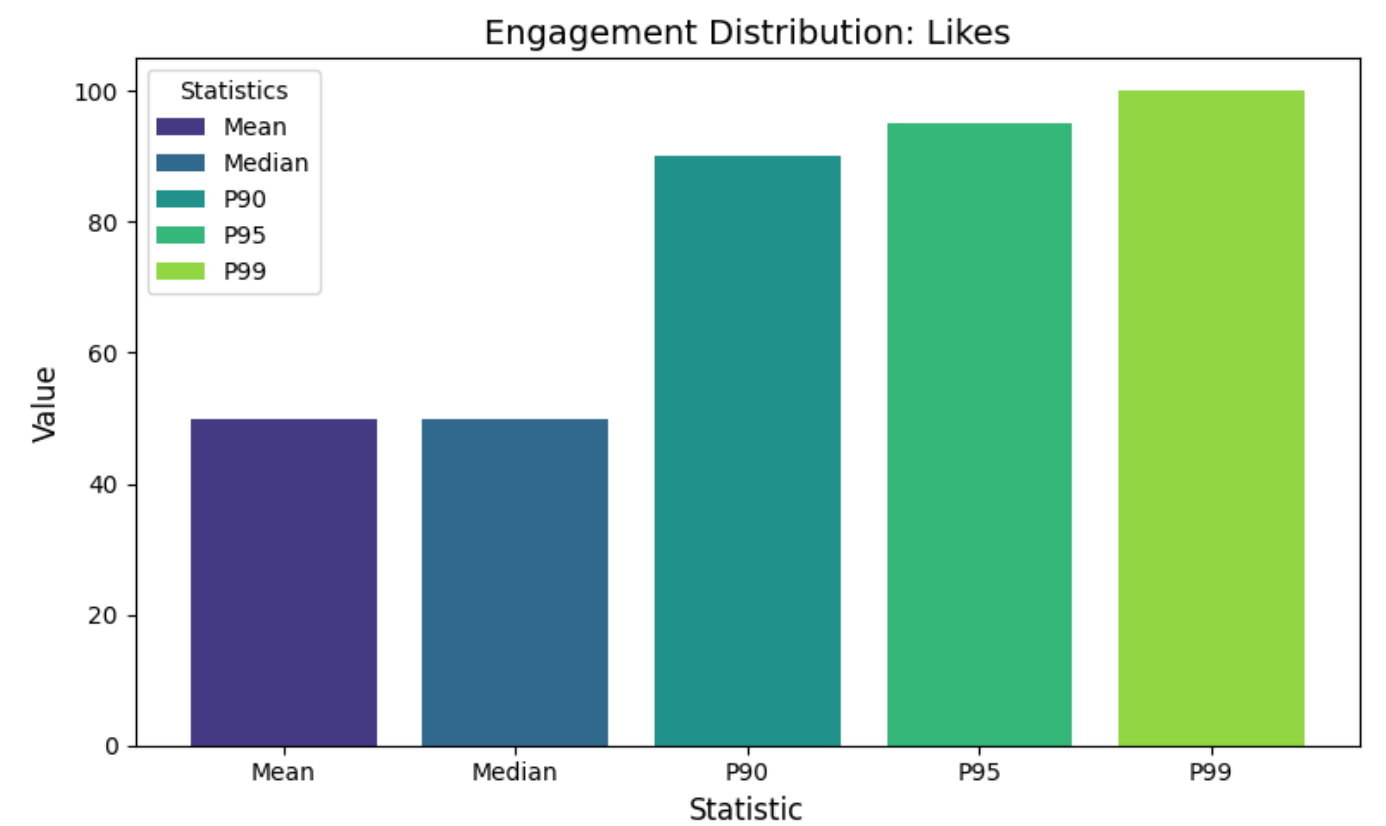
Sentiment Polarity Stats



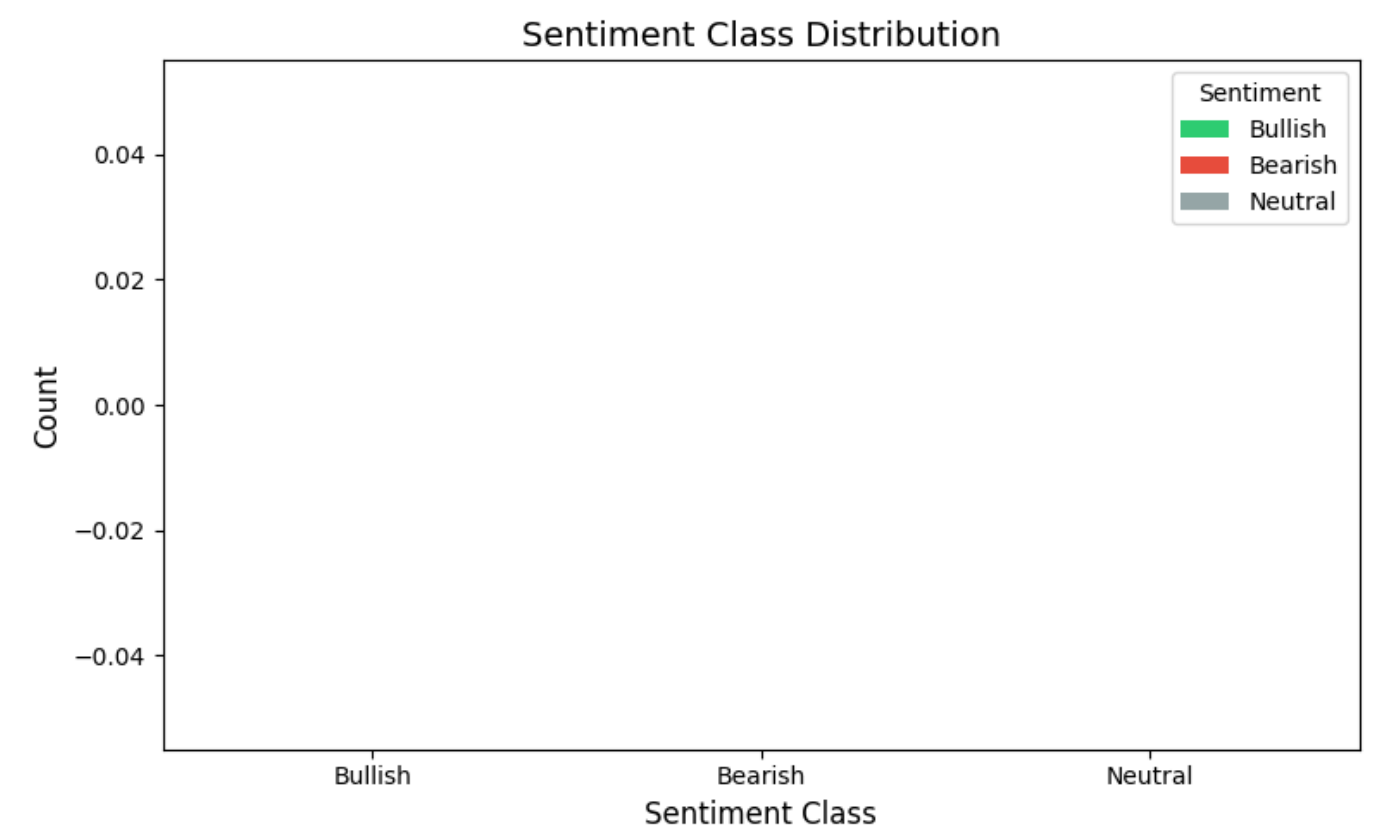
Engagement Distribution Retweets



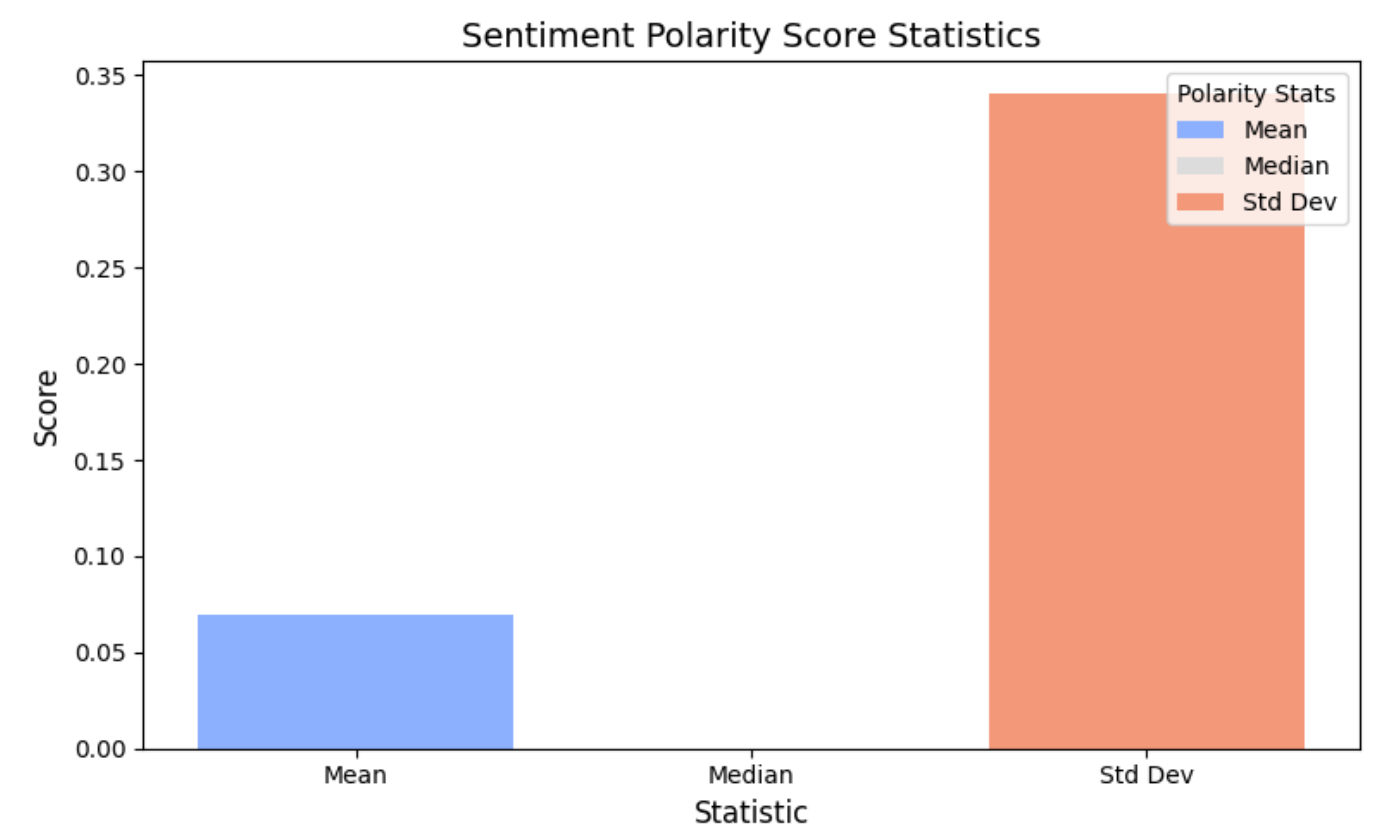
Engagement Distribution Likes



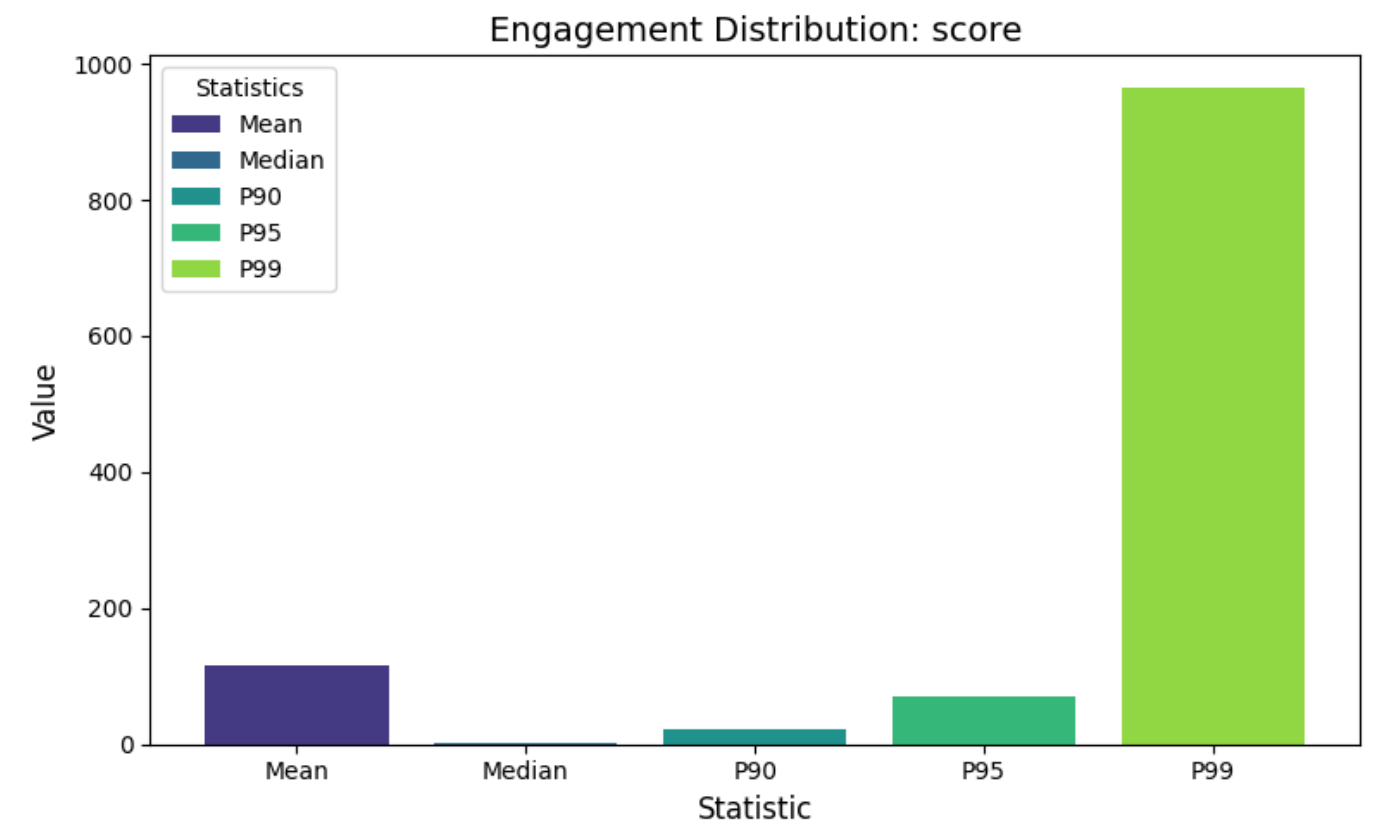
Sentiment Class Distribution



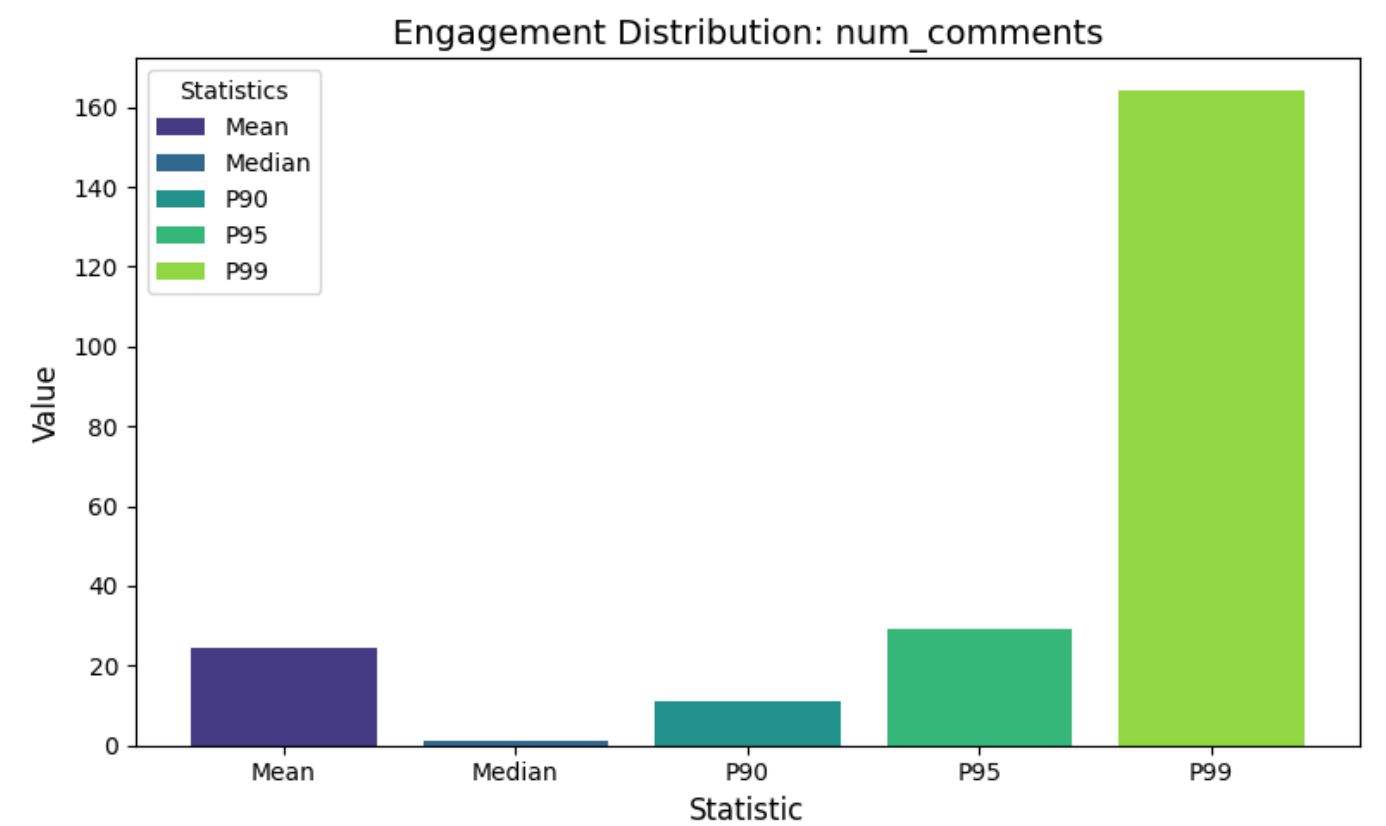
Sentiment Polarity Stats



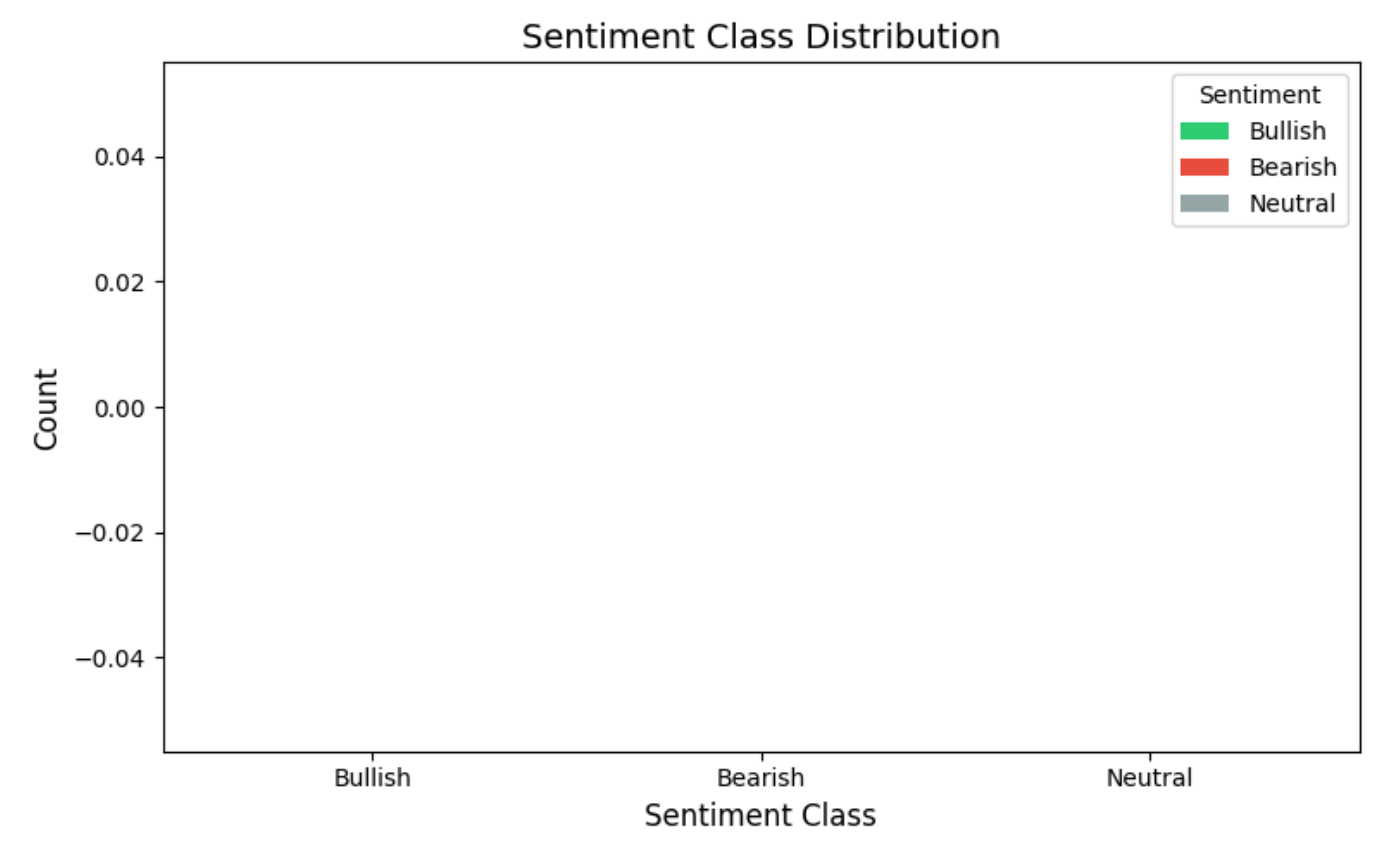
Engagement Distribution Score



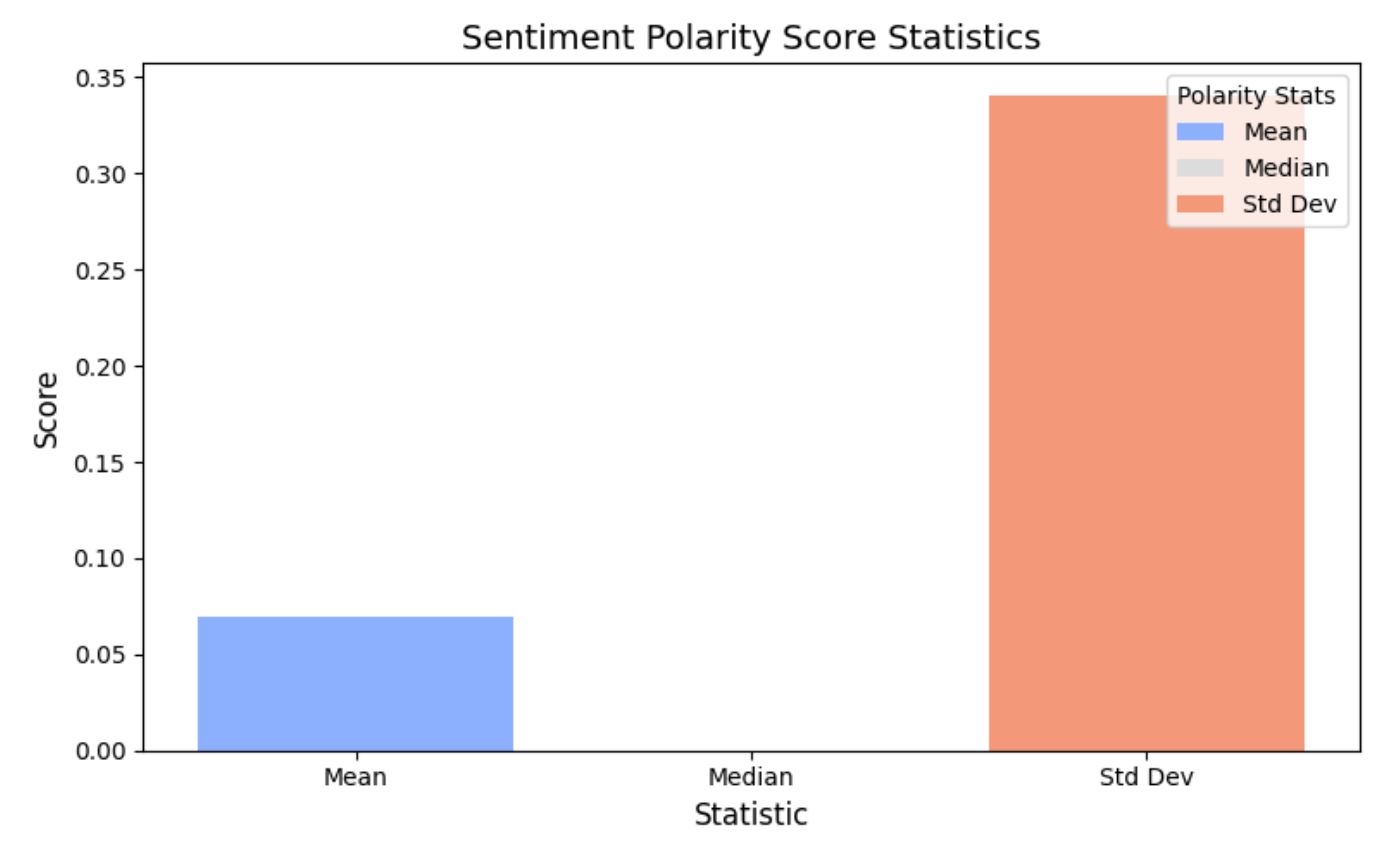
Engagement Distribution Num Comments



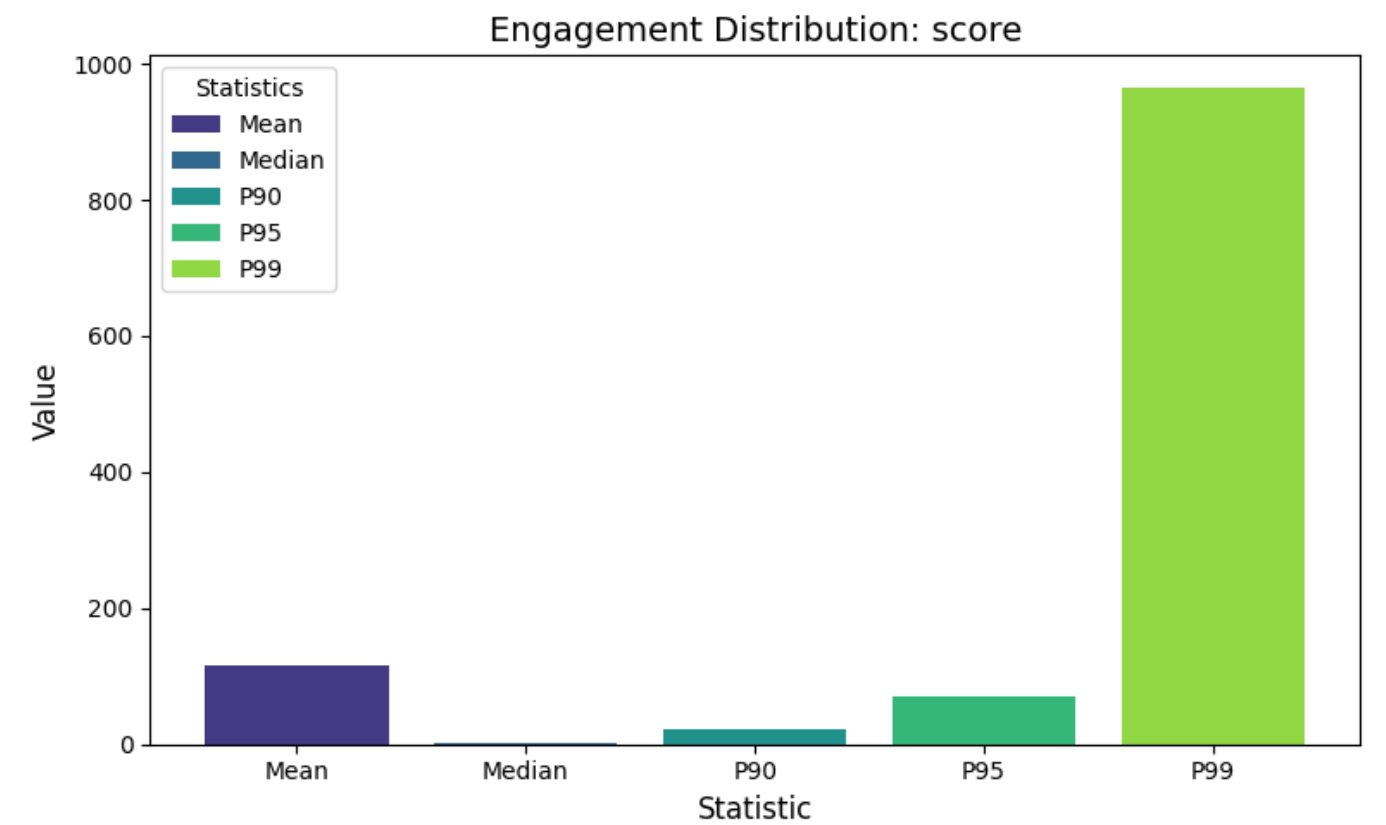
Sentiment Class Distribution



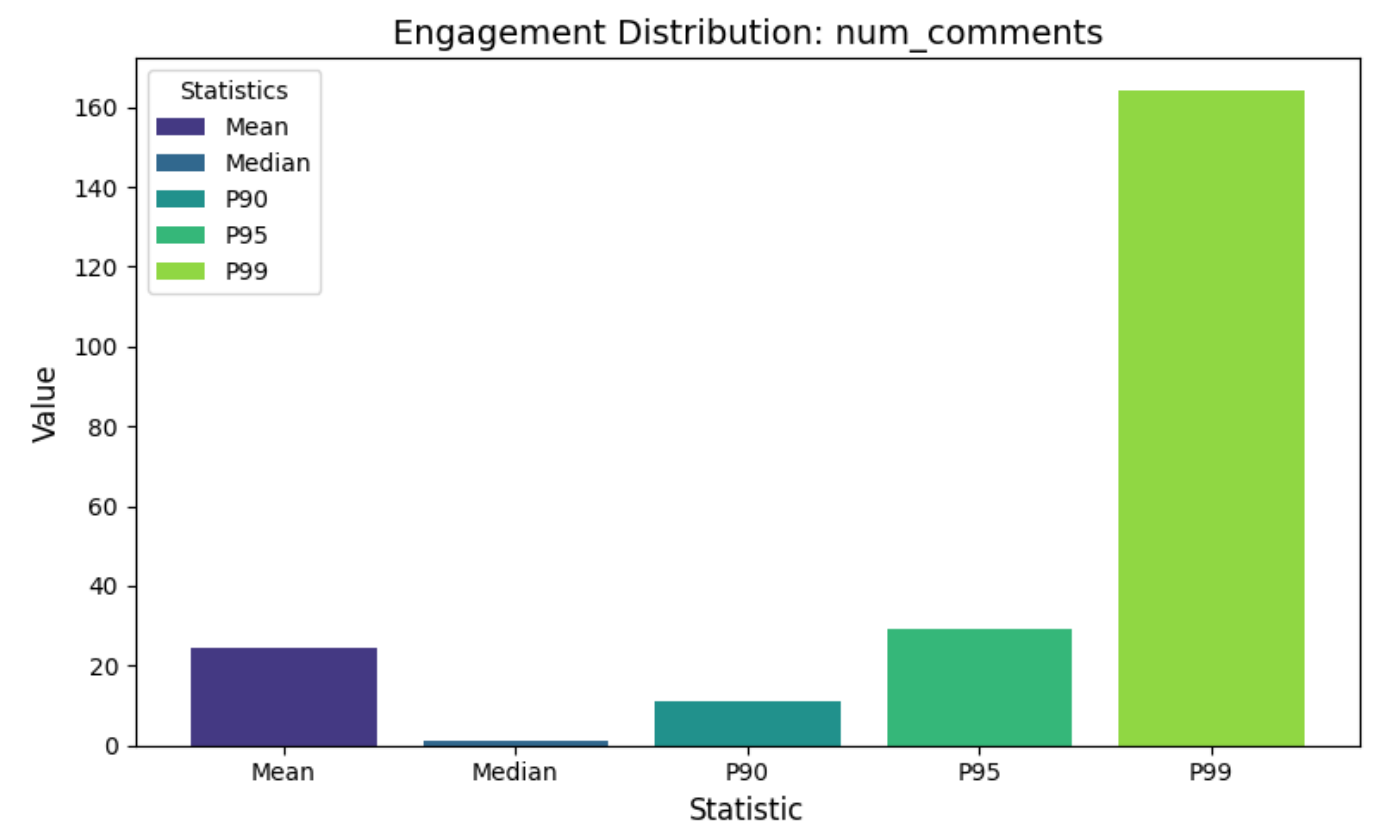
Sentiment Polarity Stats



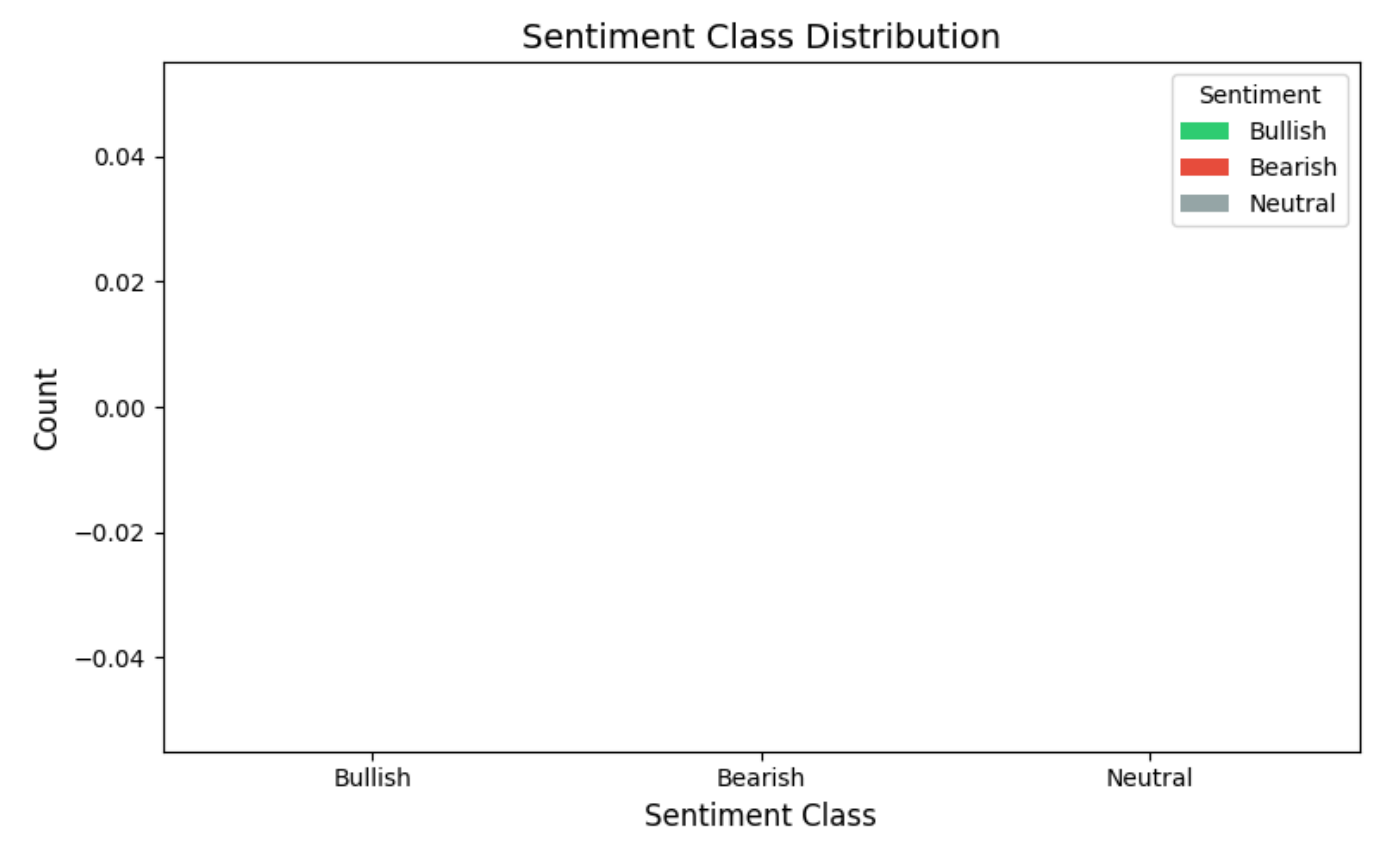
Engagement Distribution Score



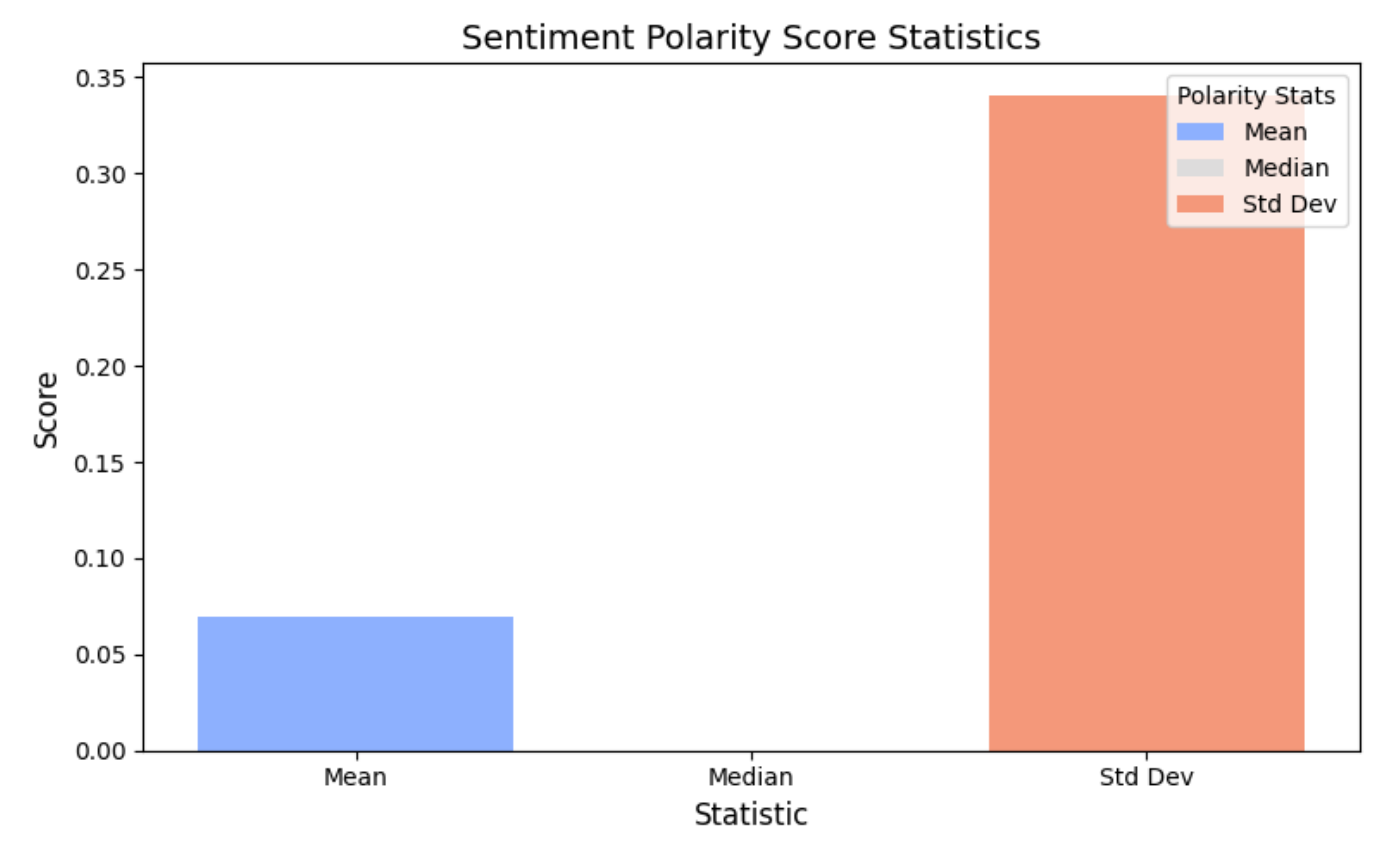
Engagement Distribution Num Comments



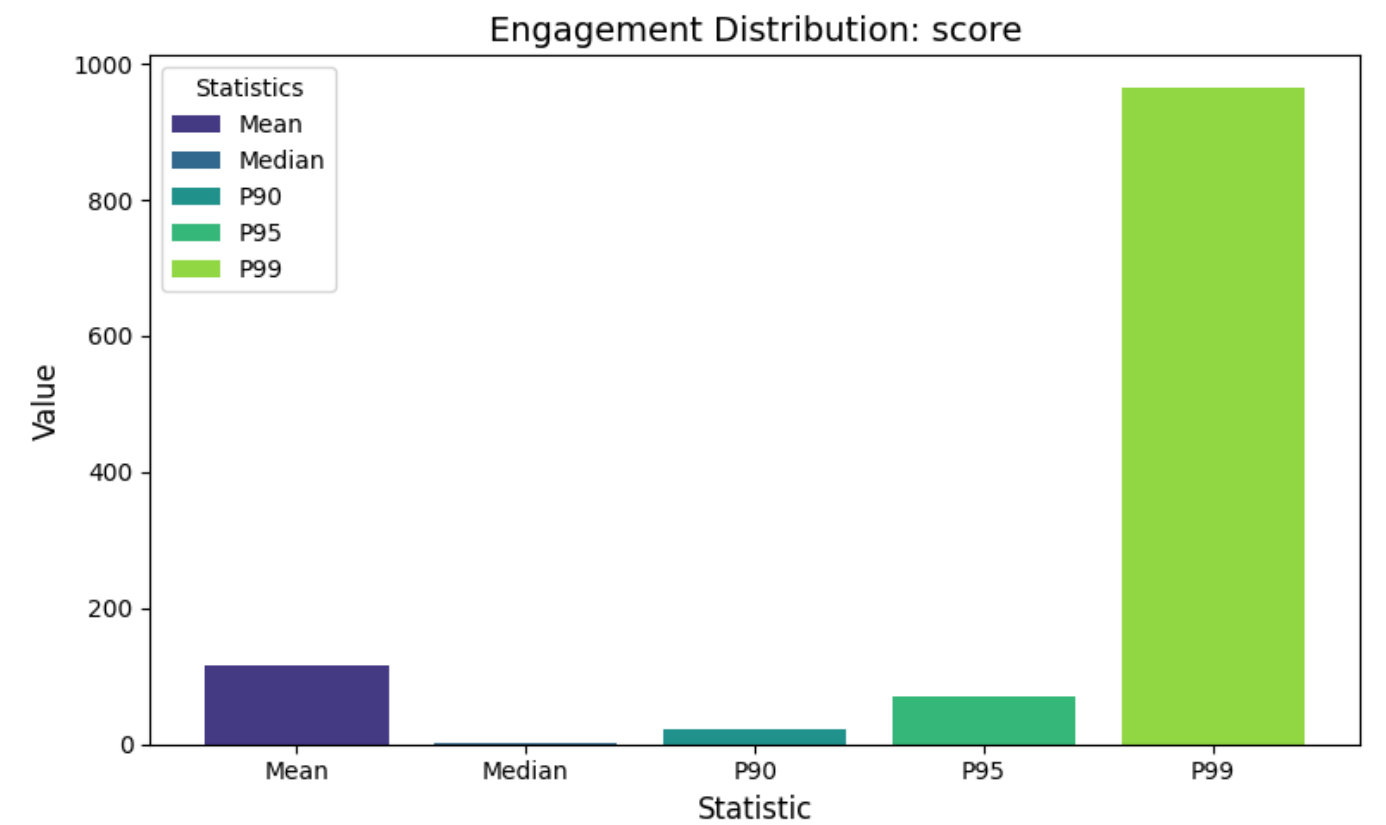
Sentiment Class Distribution



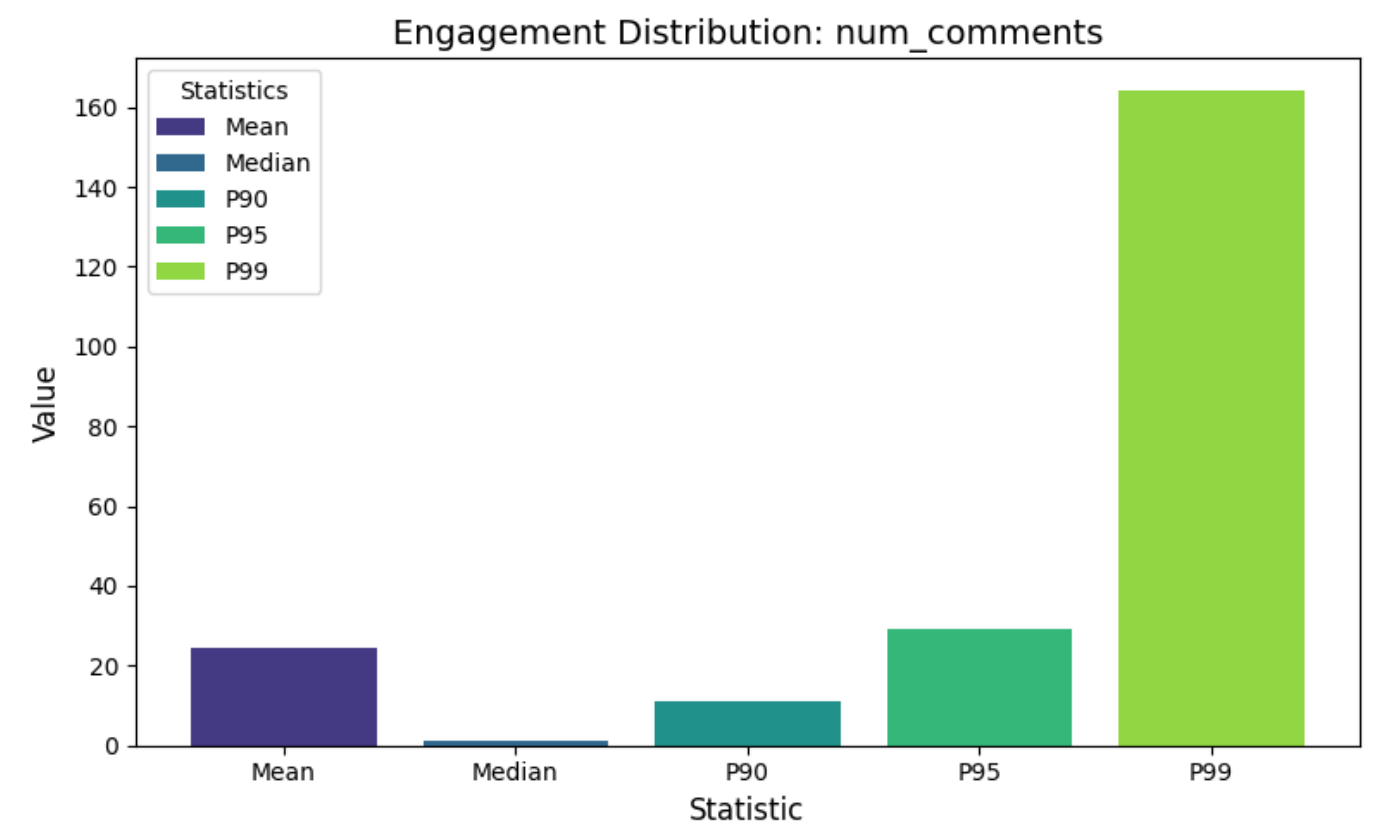
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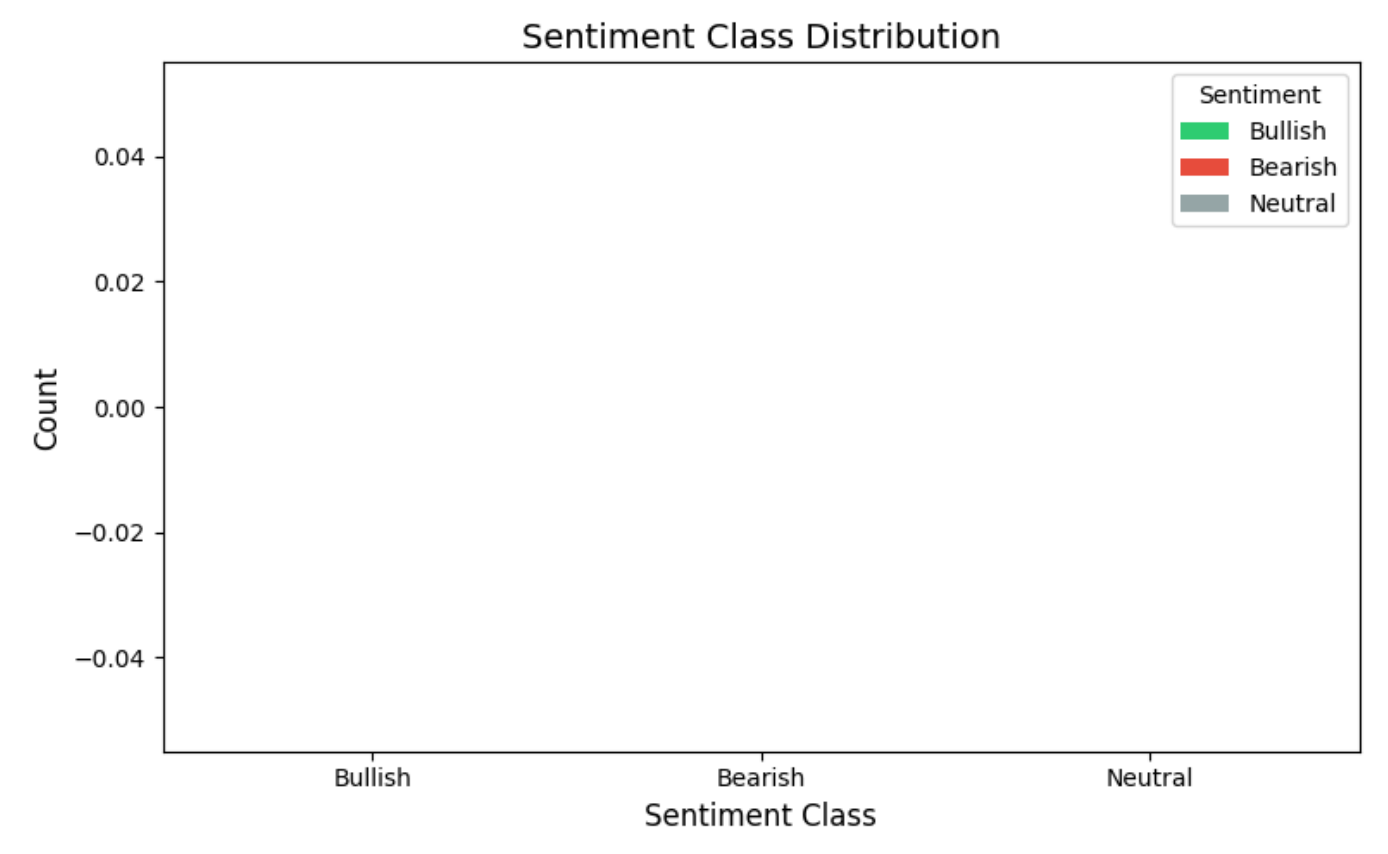
Engagement Distribution Score



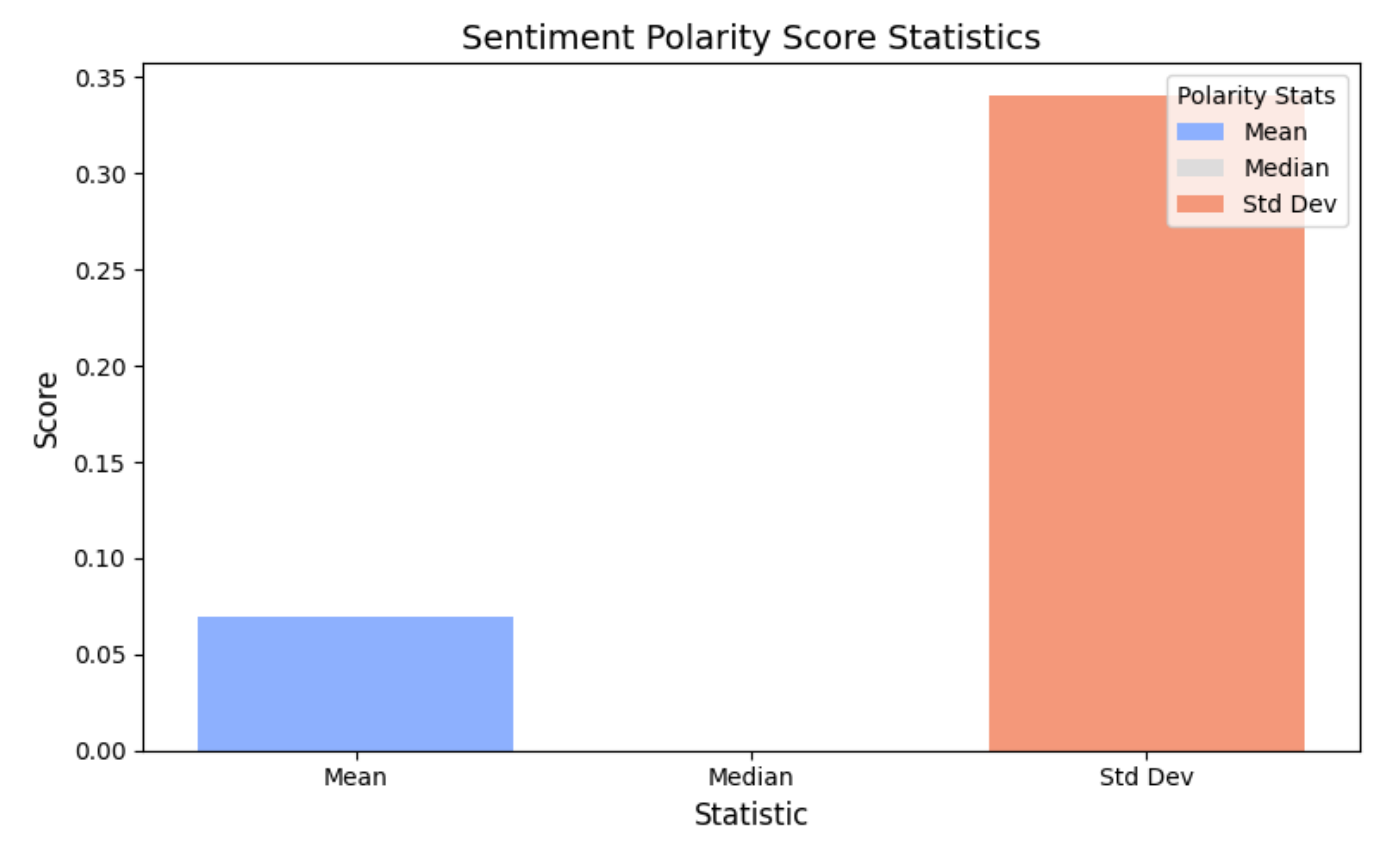
Engagement Distribution Num Comments



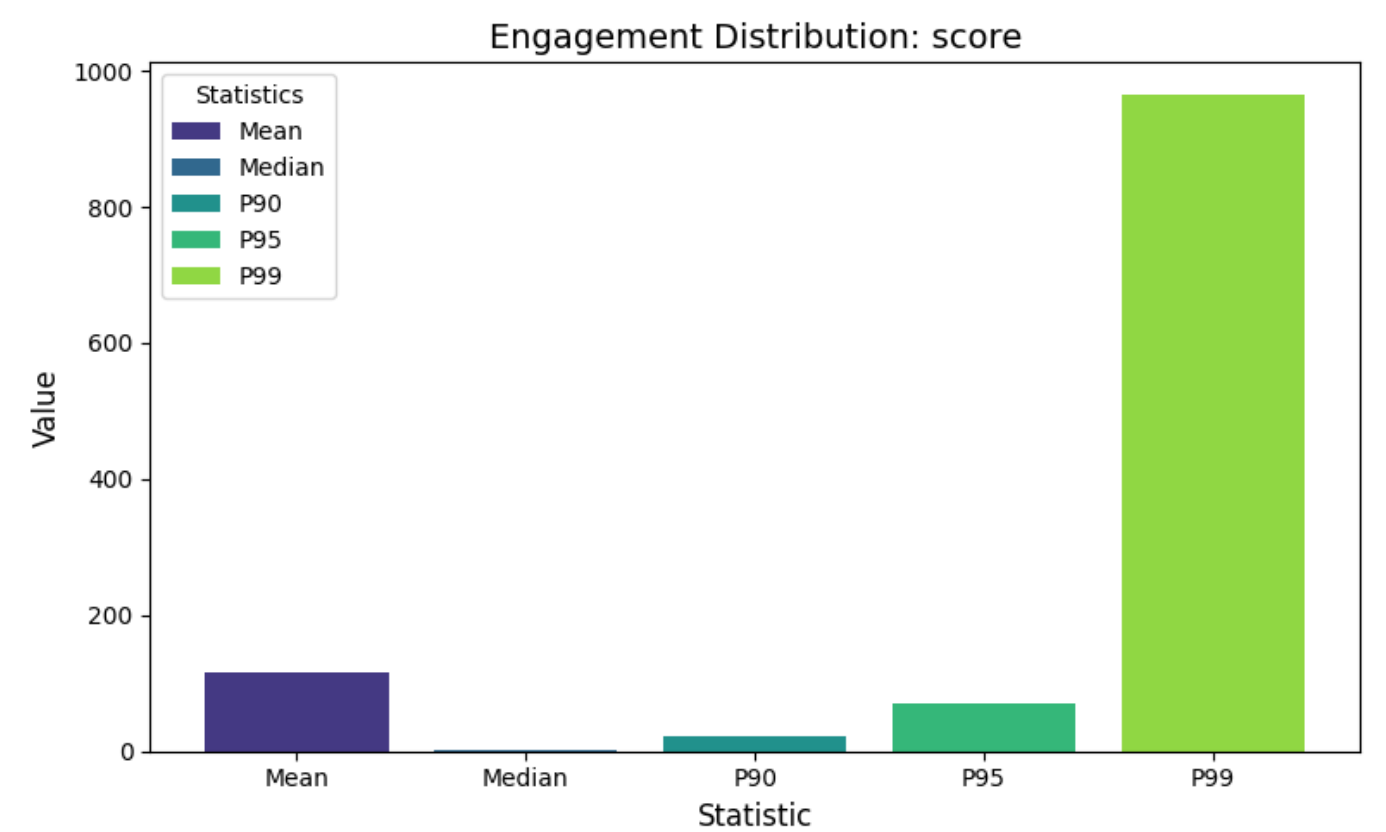
Sentiment Class Distribution



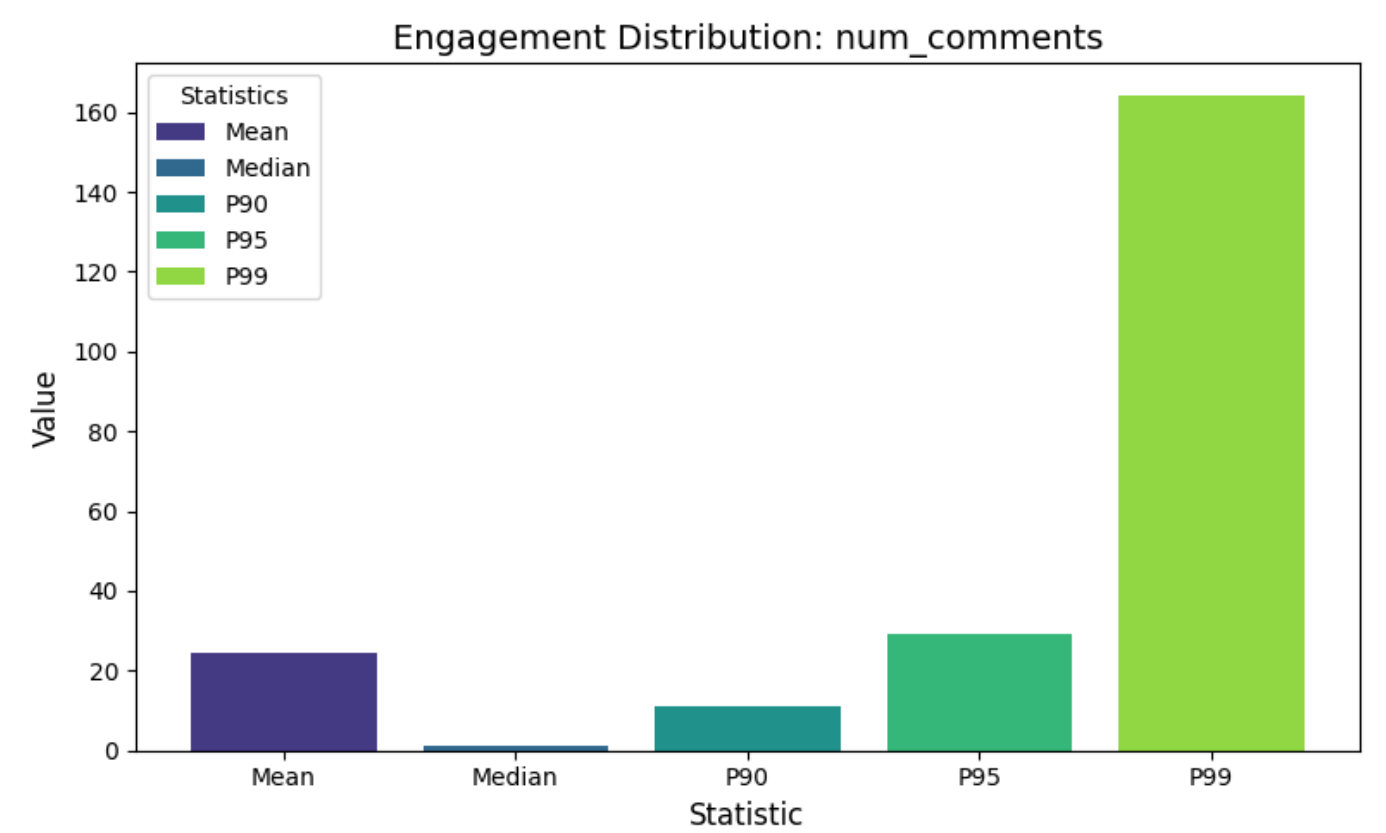
Sentiment Polarity Stats



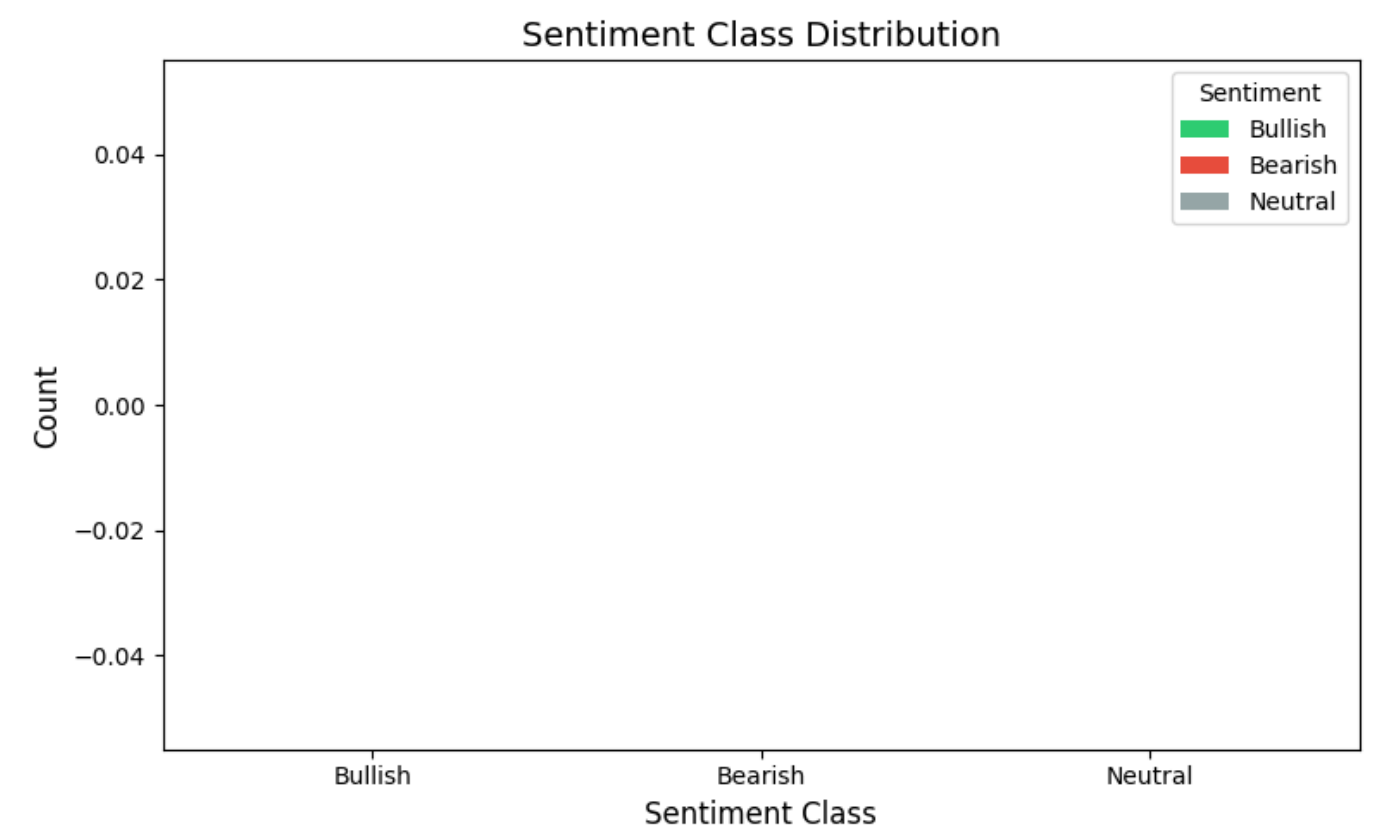
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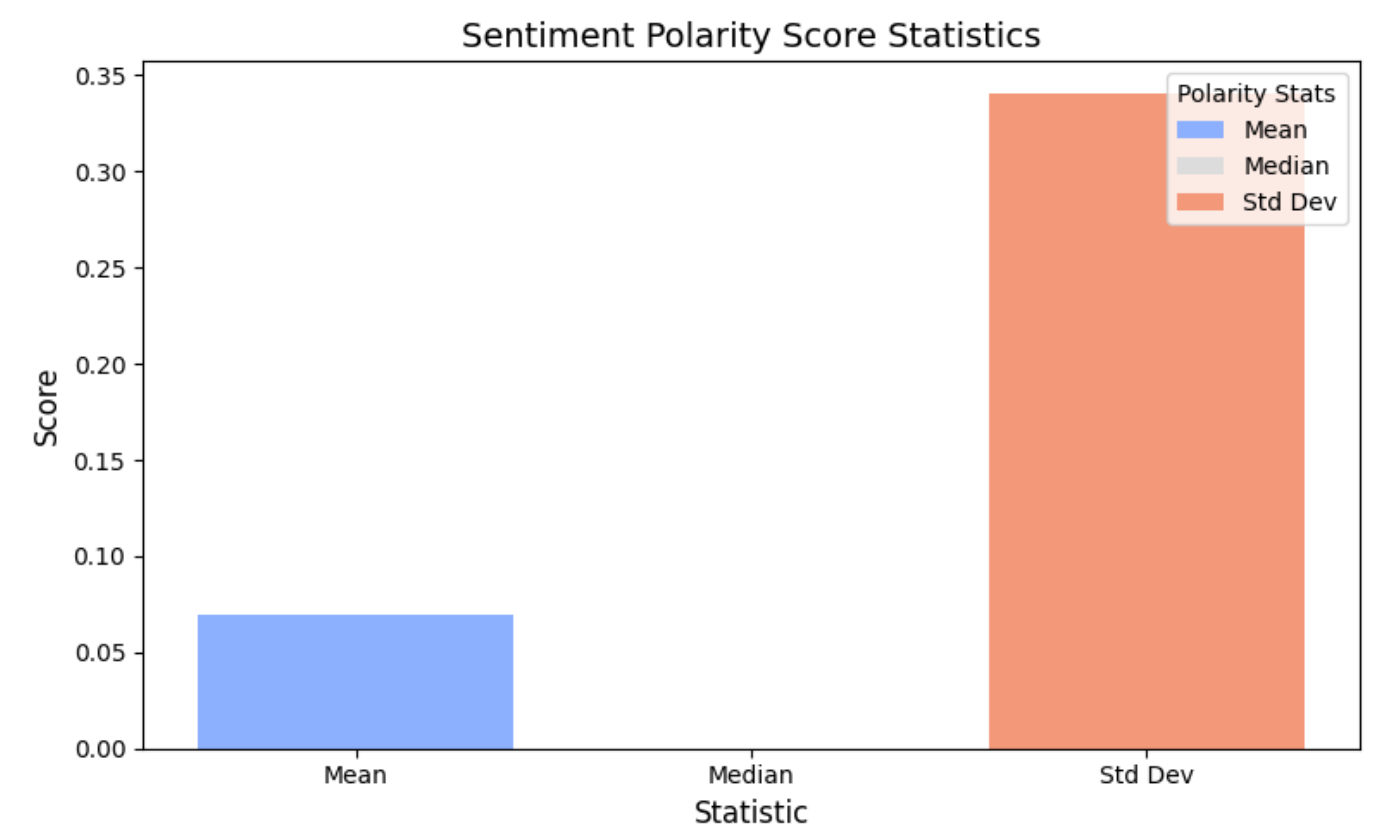
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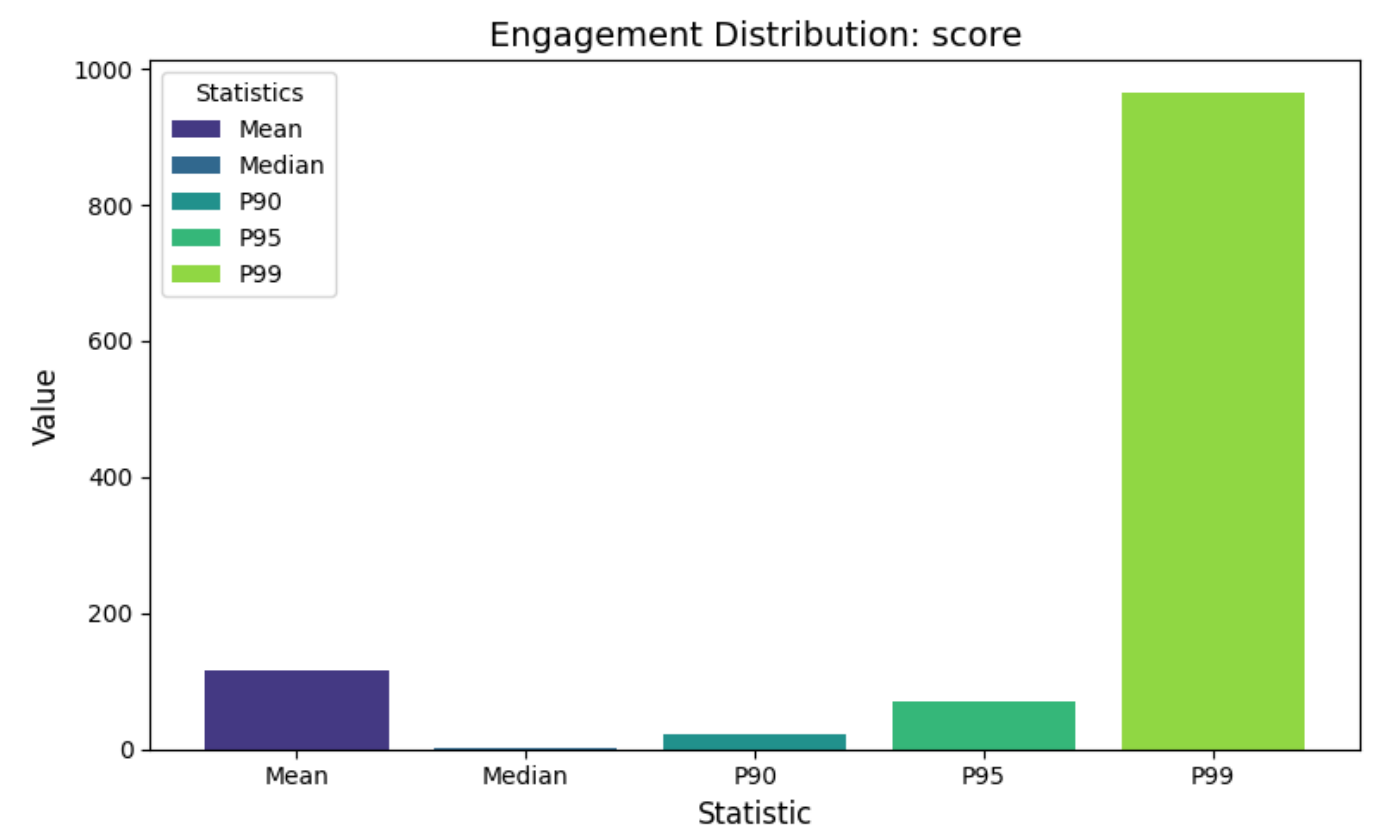
Sentiment Class Distribution



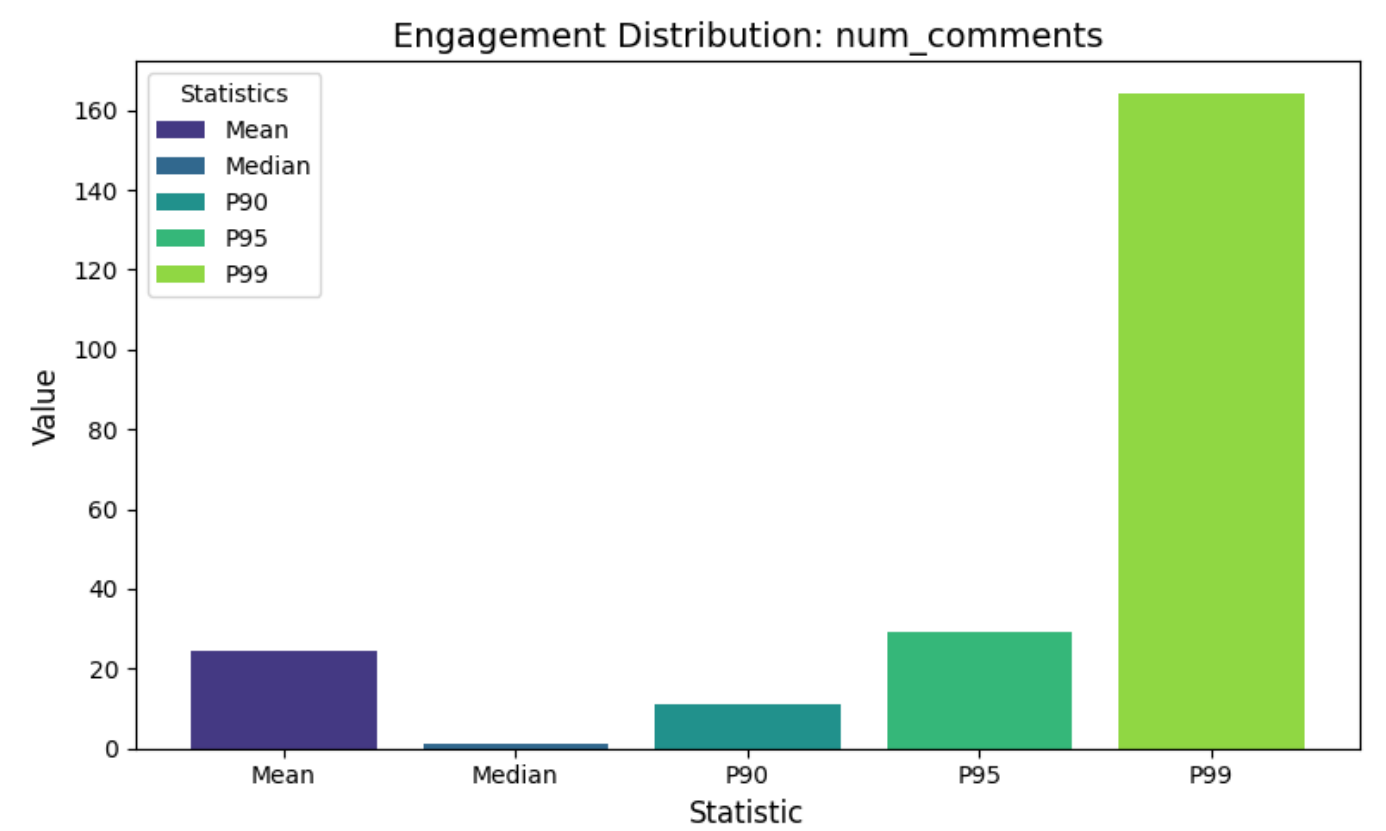
Sentiment Polarity Stats



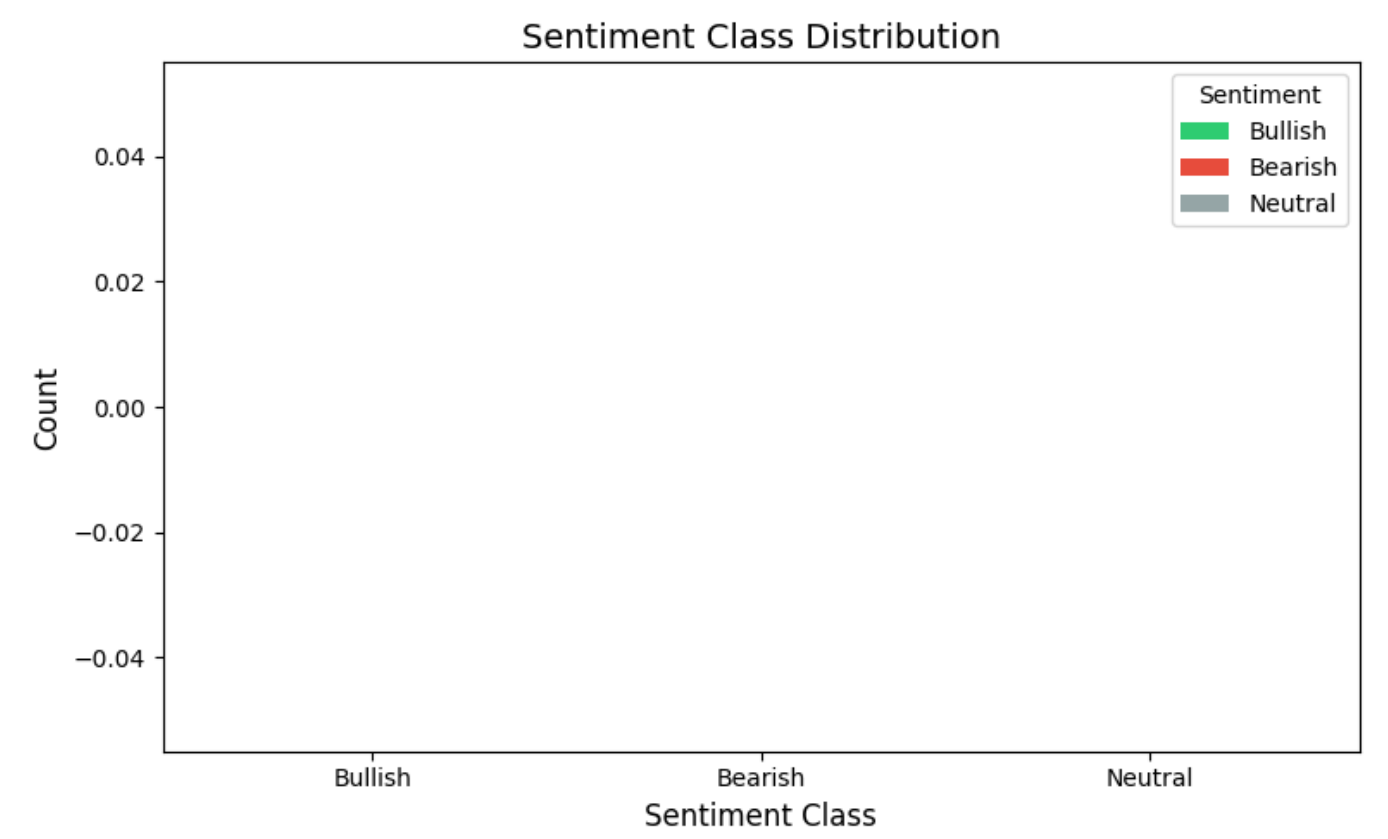
Engagement Distribution Score



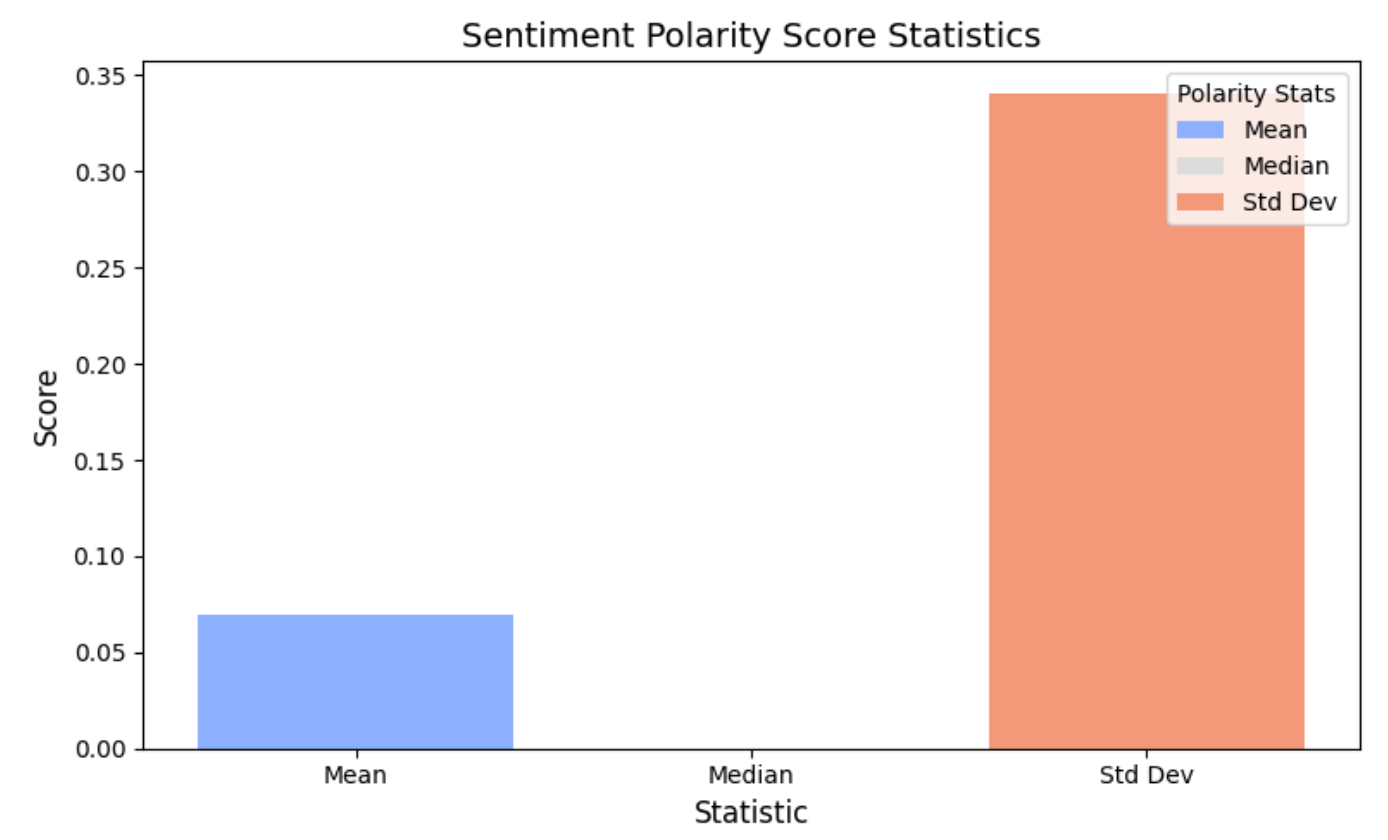
Engagement Distribution Num Comments



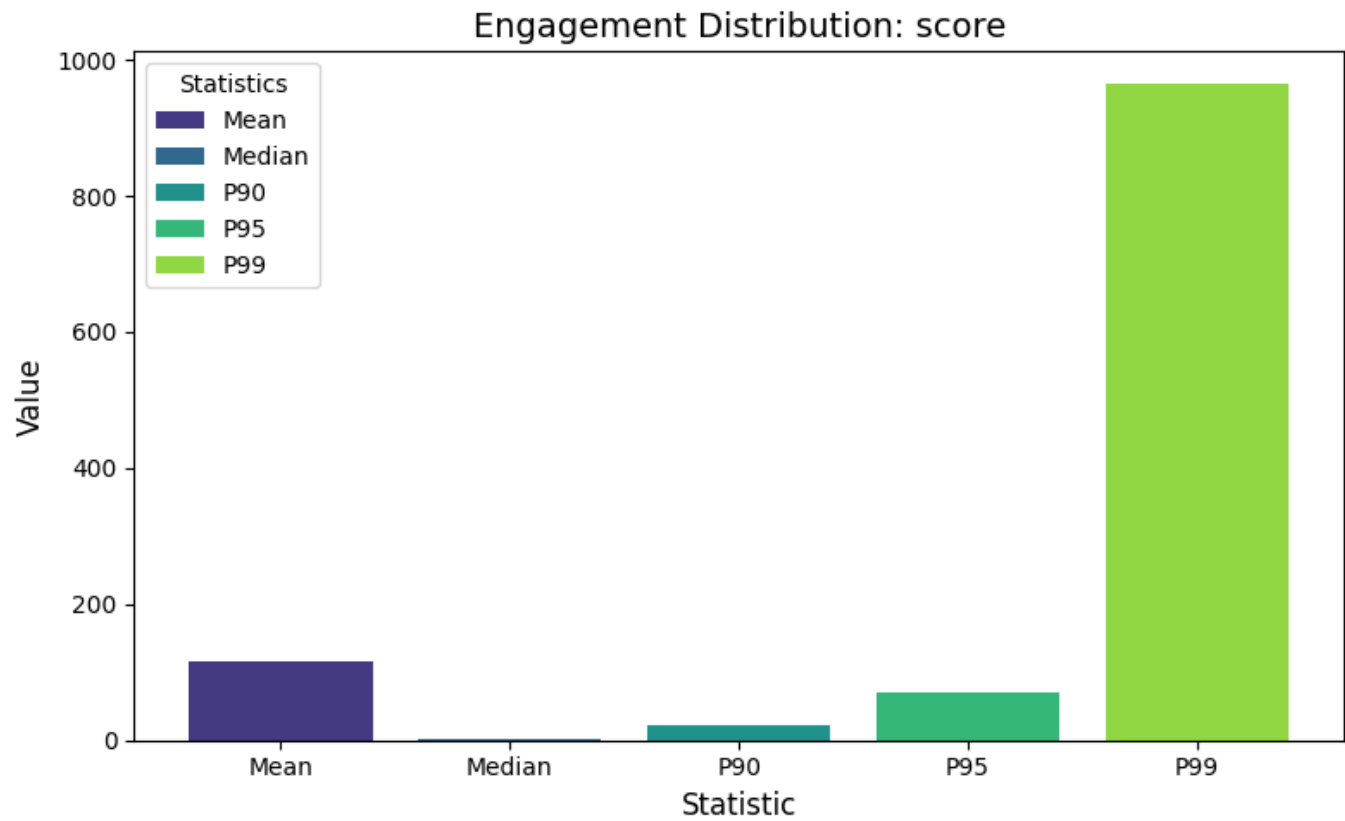
Sentiment Class Distribution



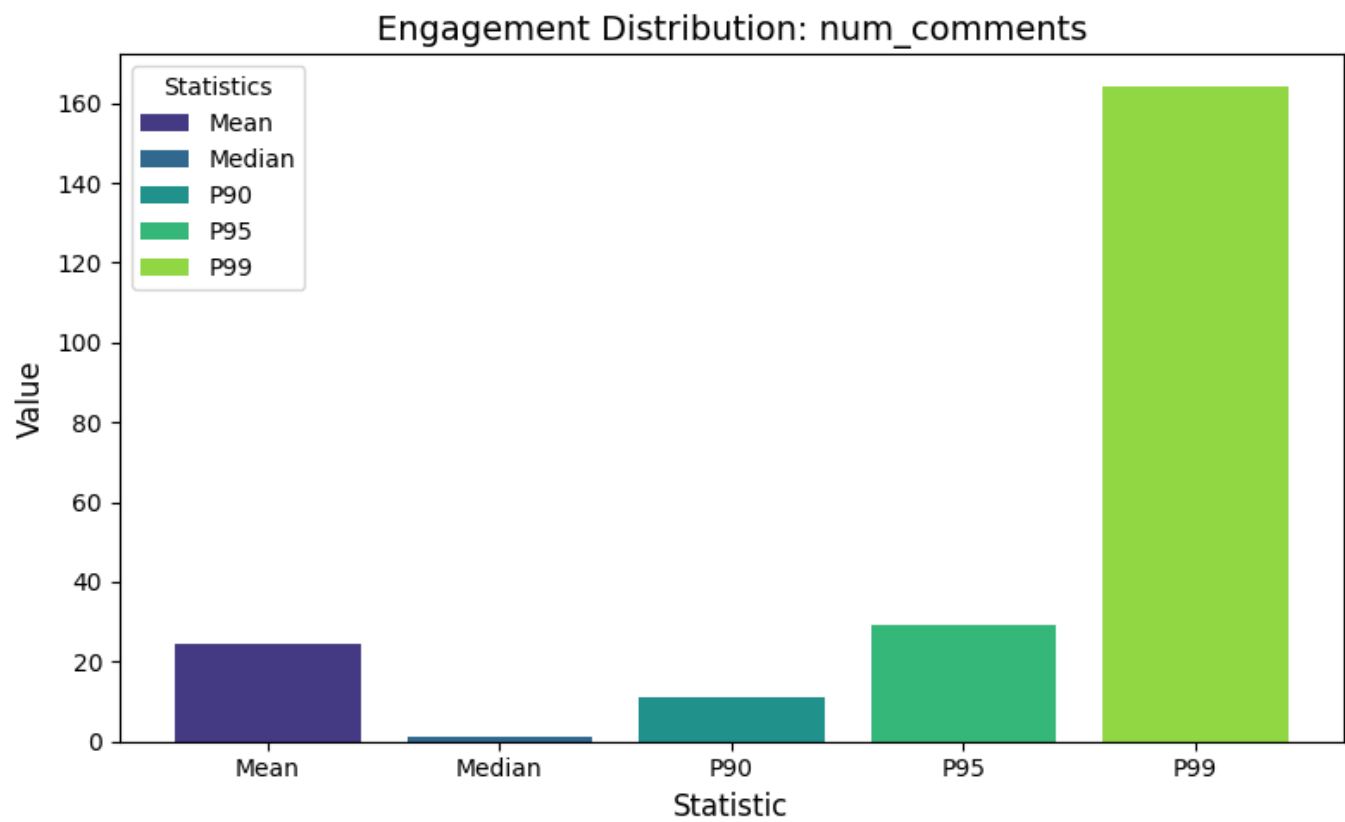
Sentiment Polarity Stats



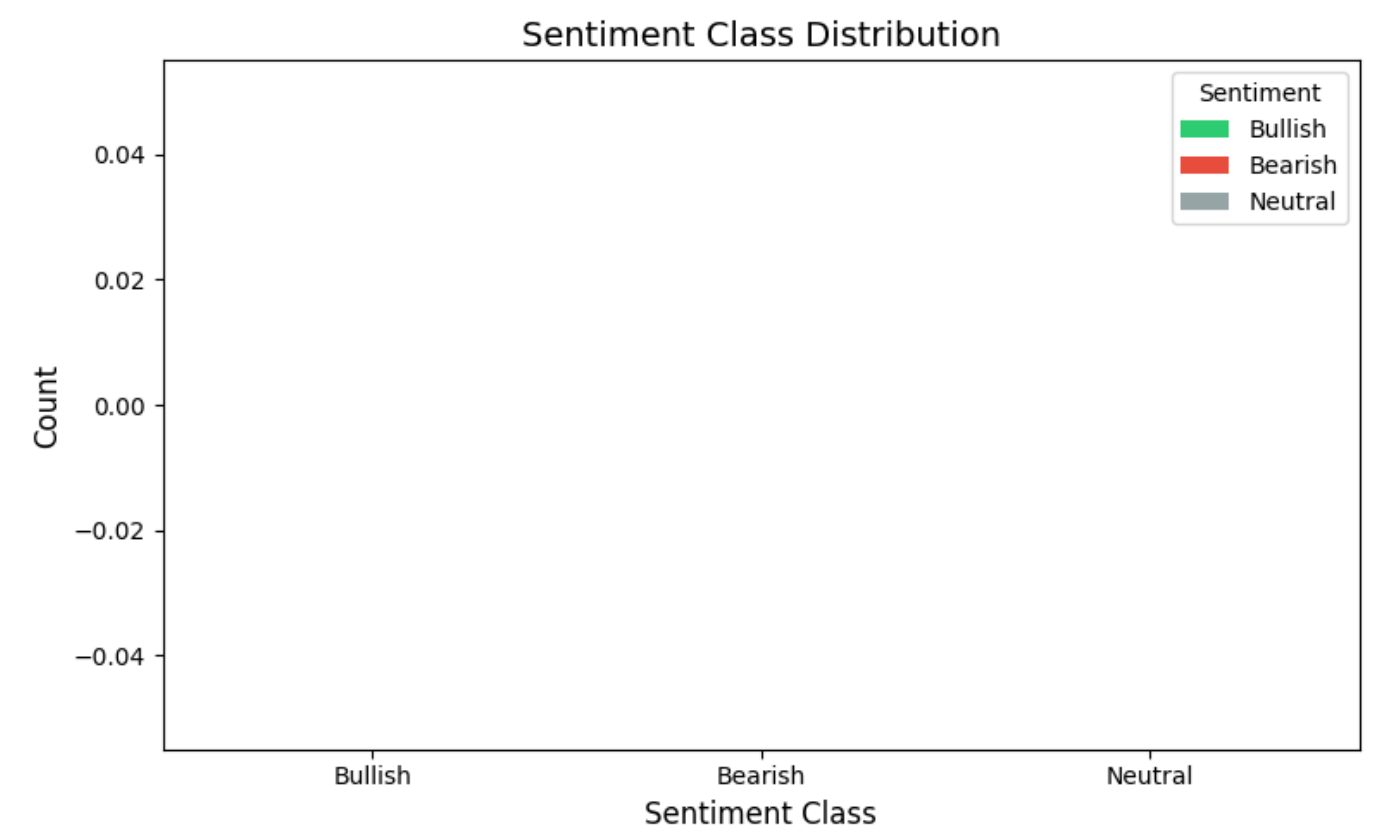
Engagement Distribution Score



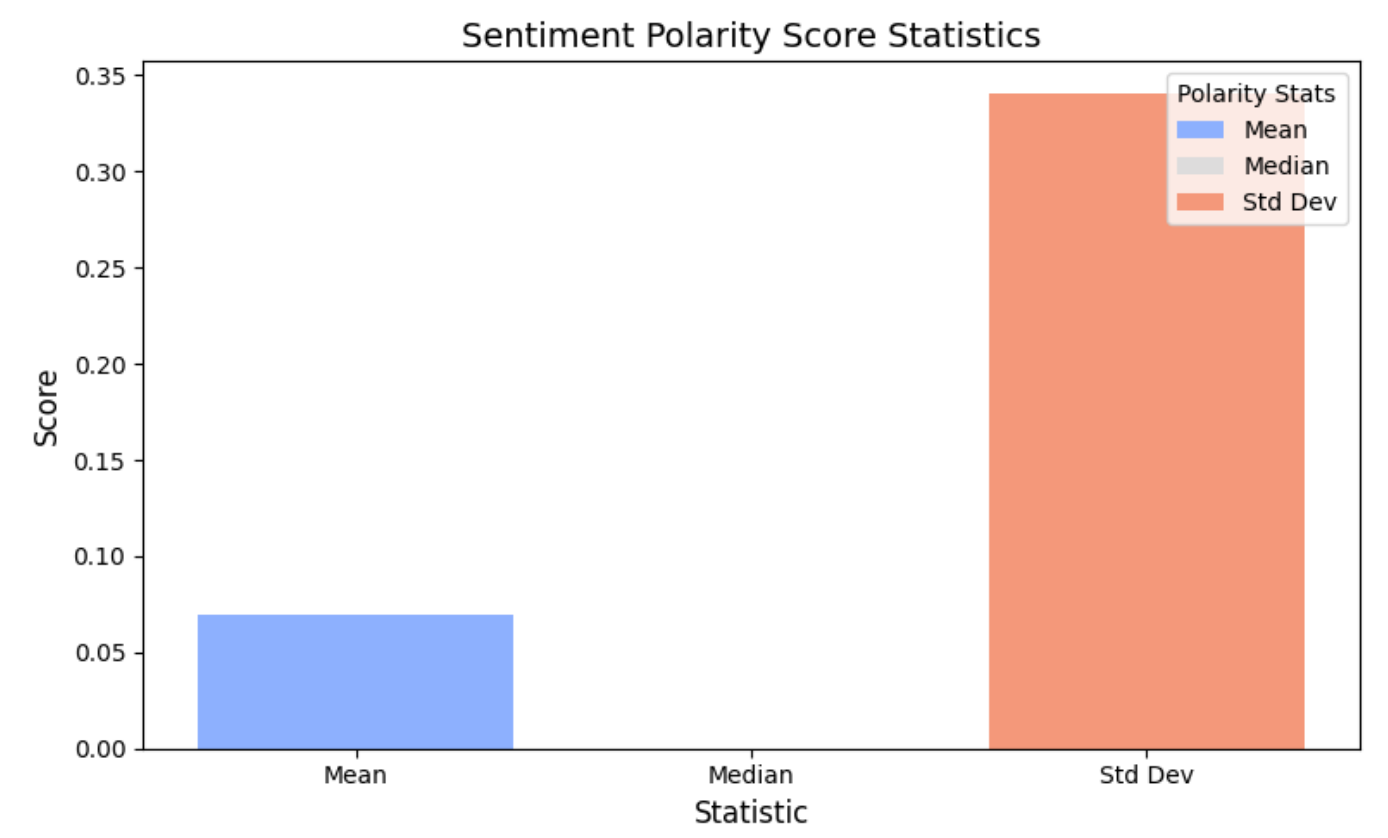
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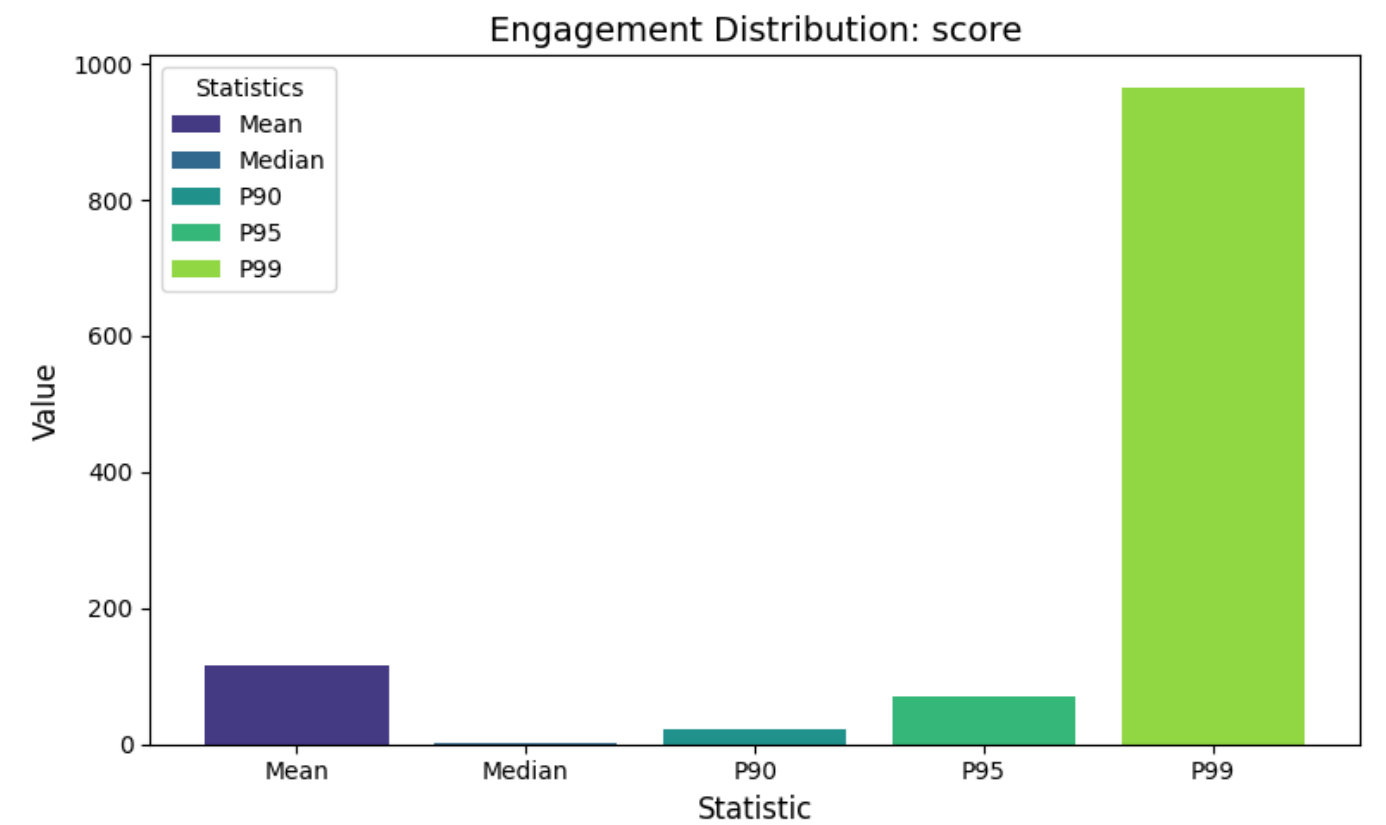
Sentiment Class Distribution



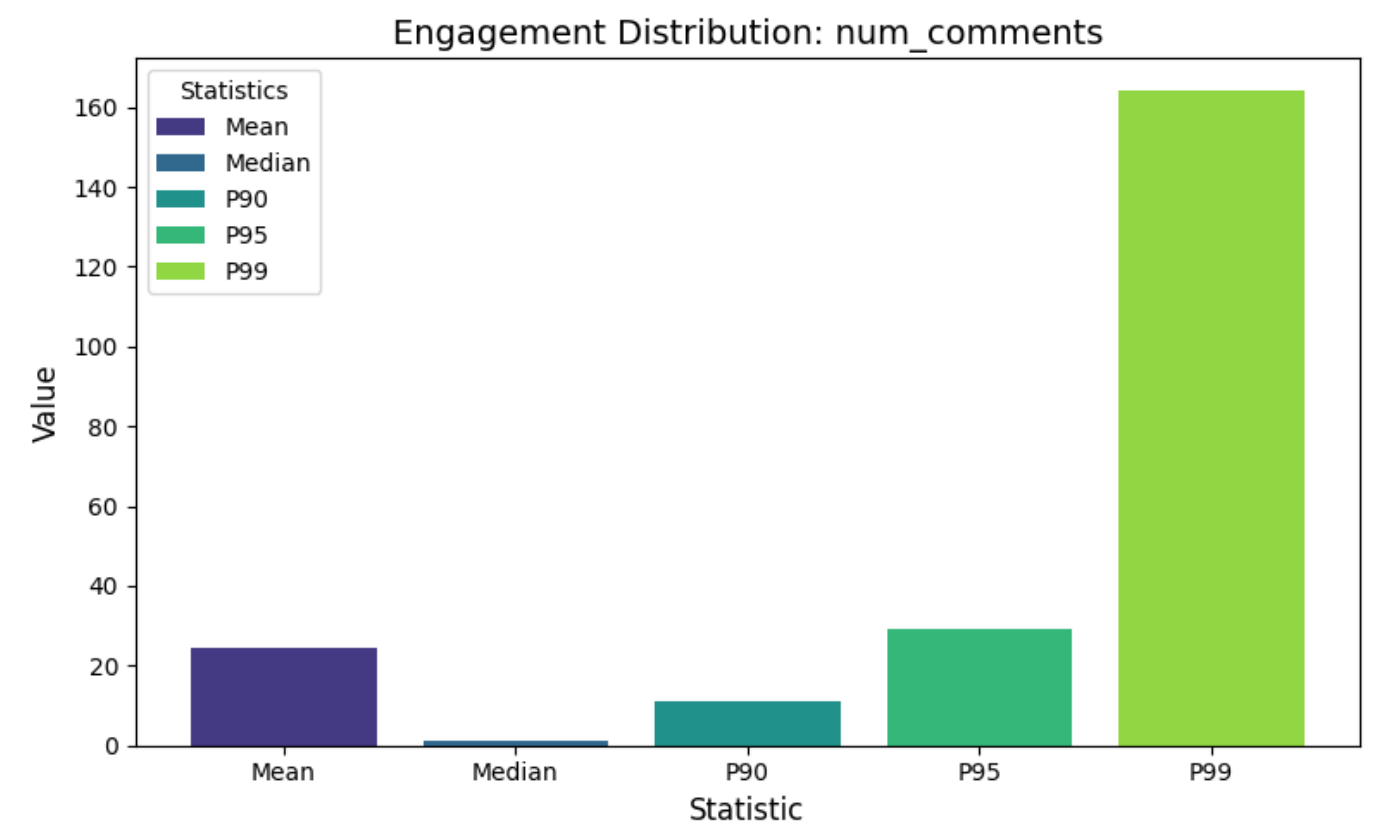
Sentiment Polarity Stats



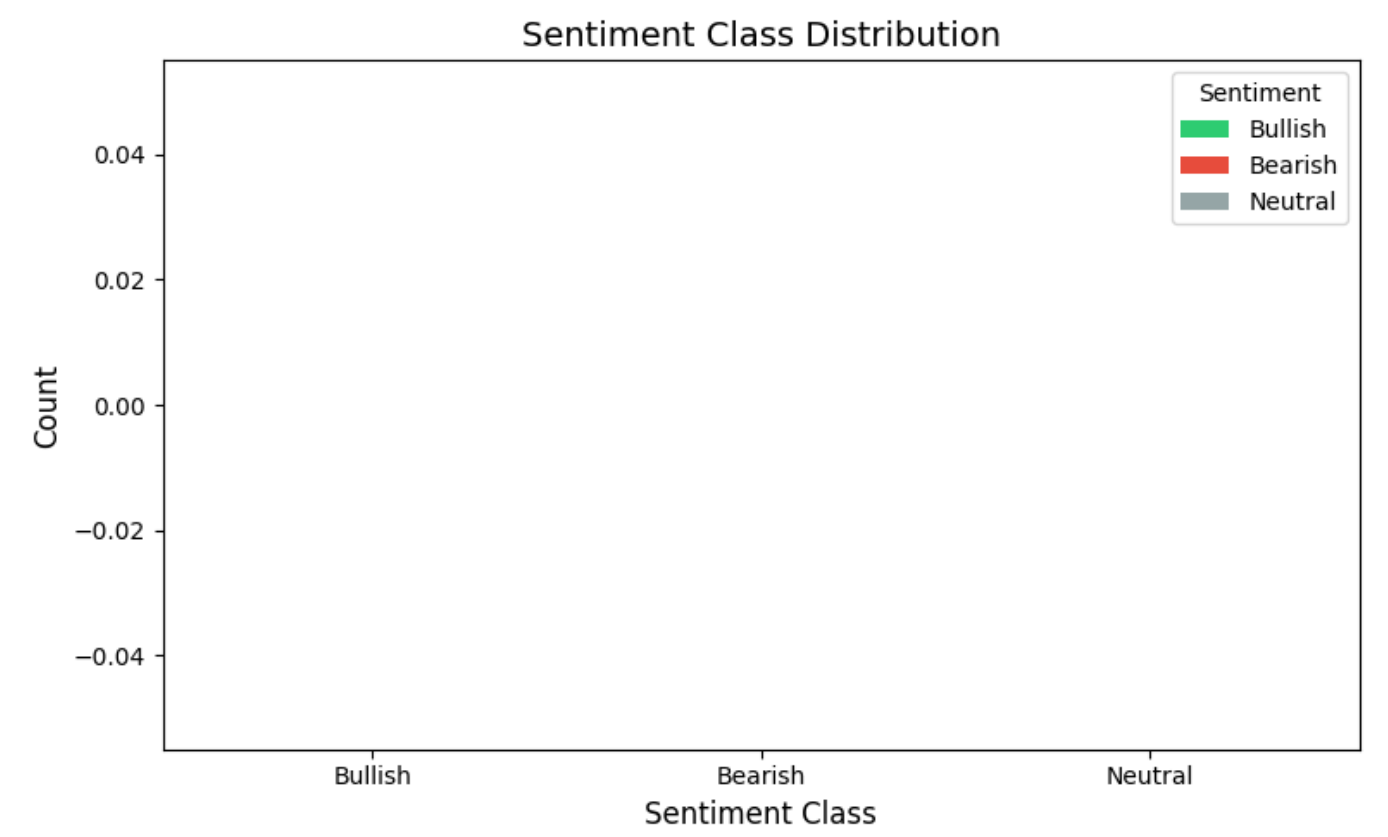
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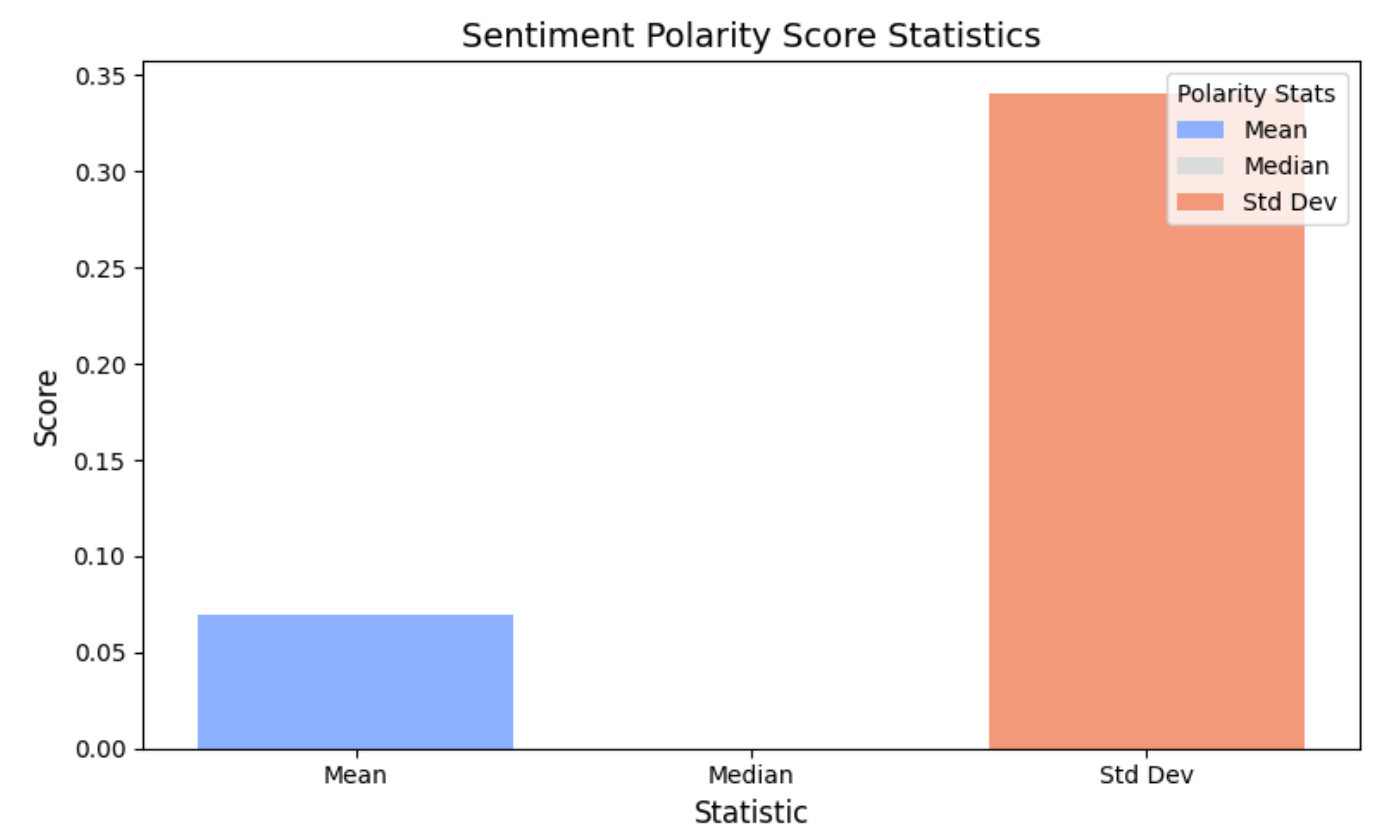
Engagement Distribution Num Comments



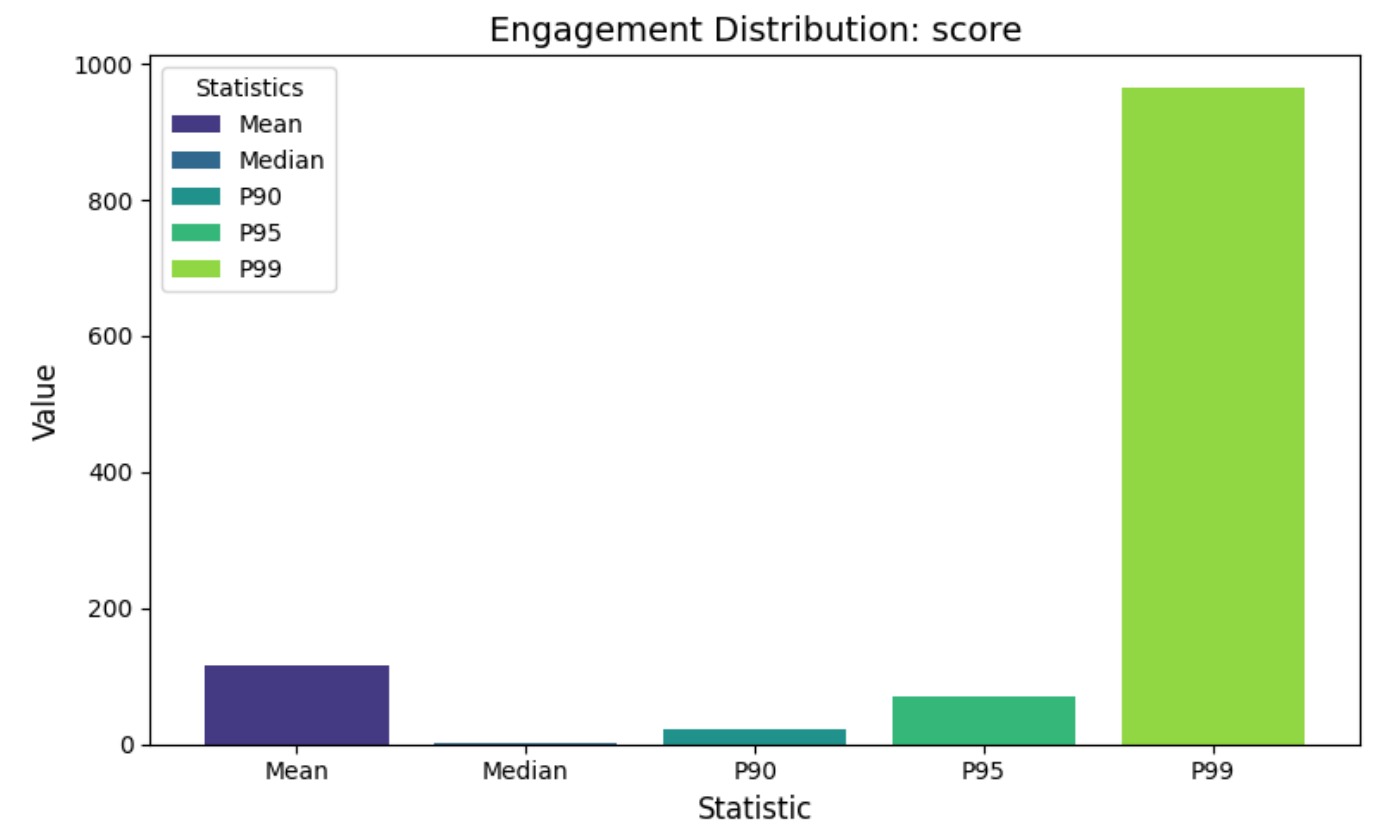
Sentiment Class Distribution



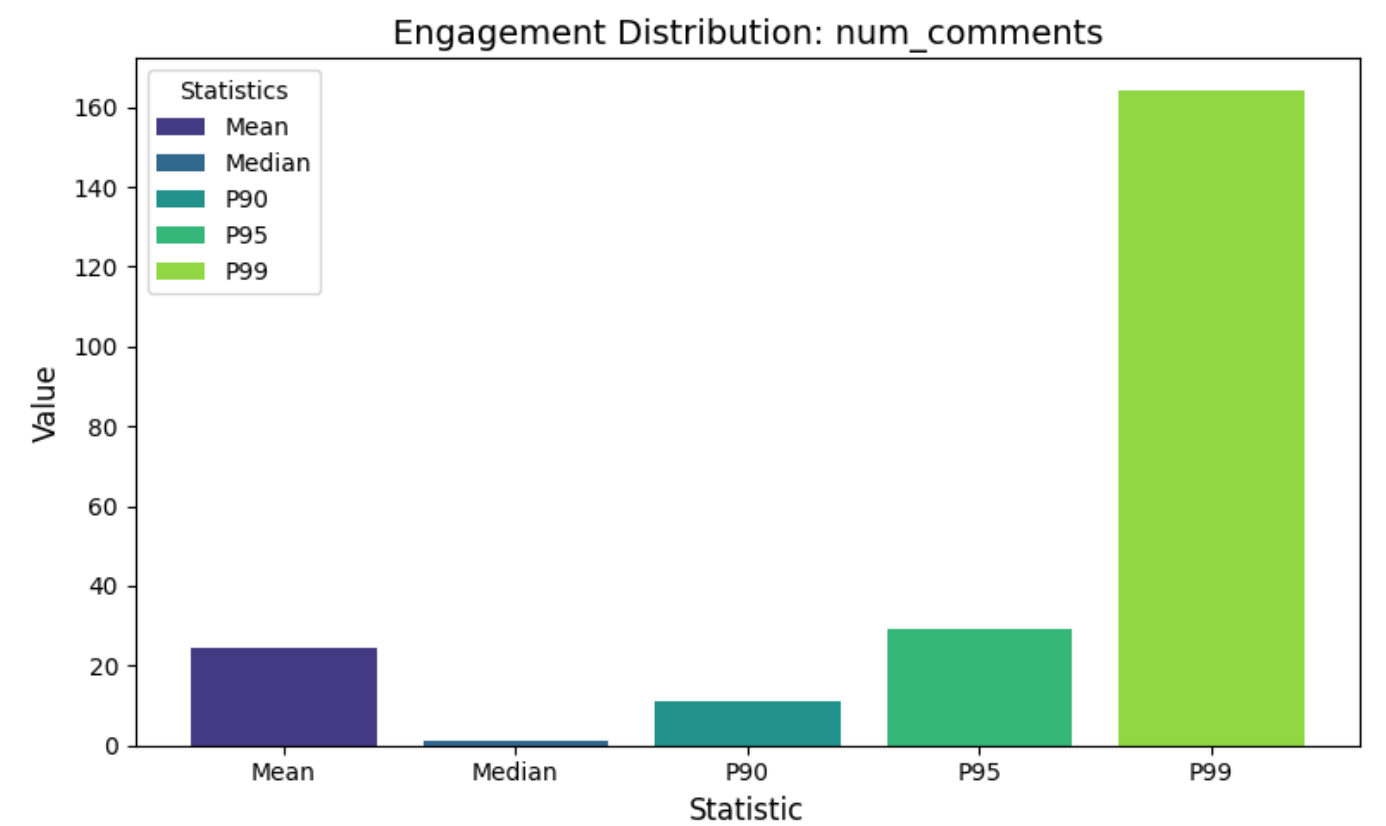
Sentiment Polarity Stats



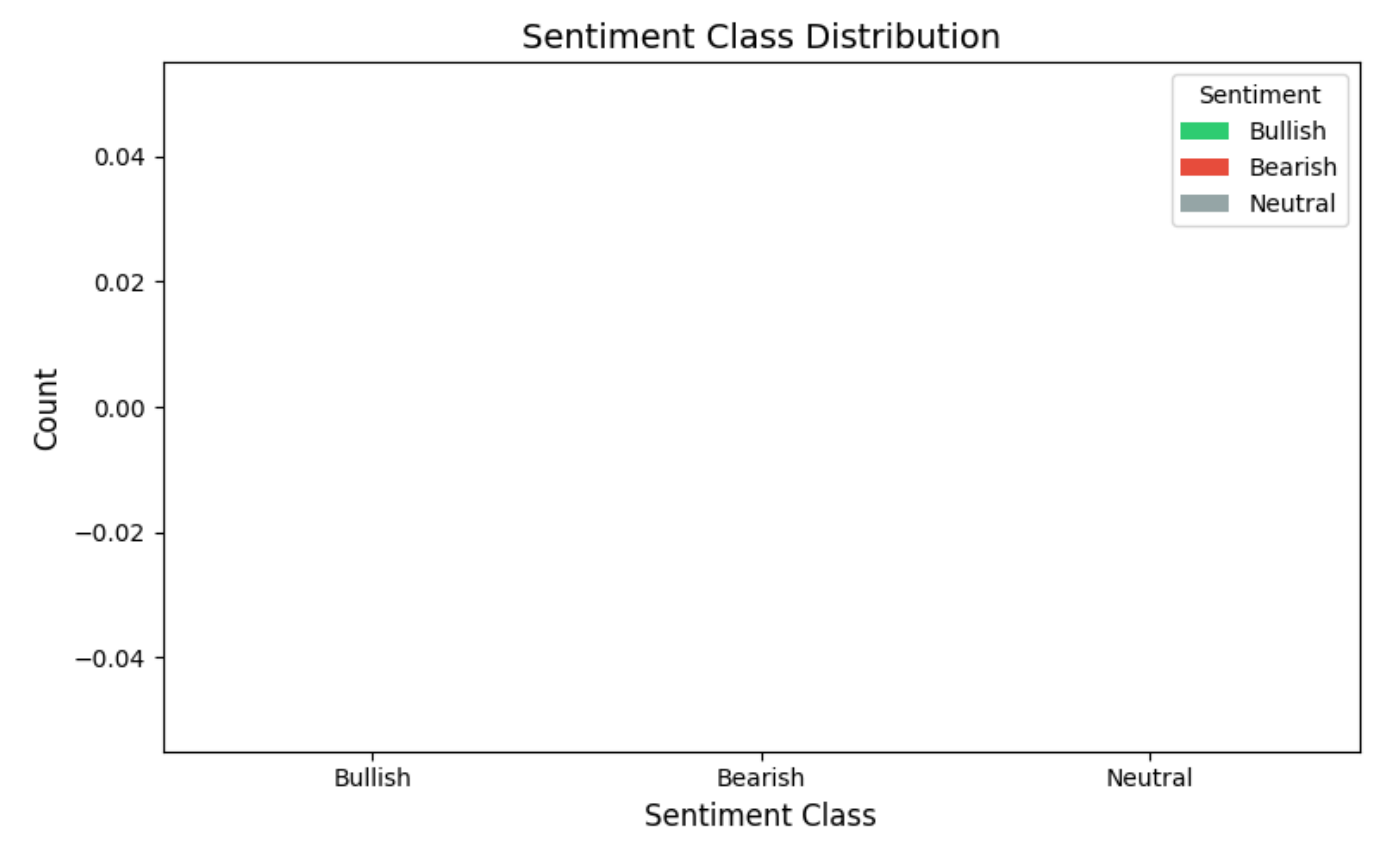
Engagement Distribution Score



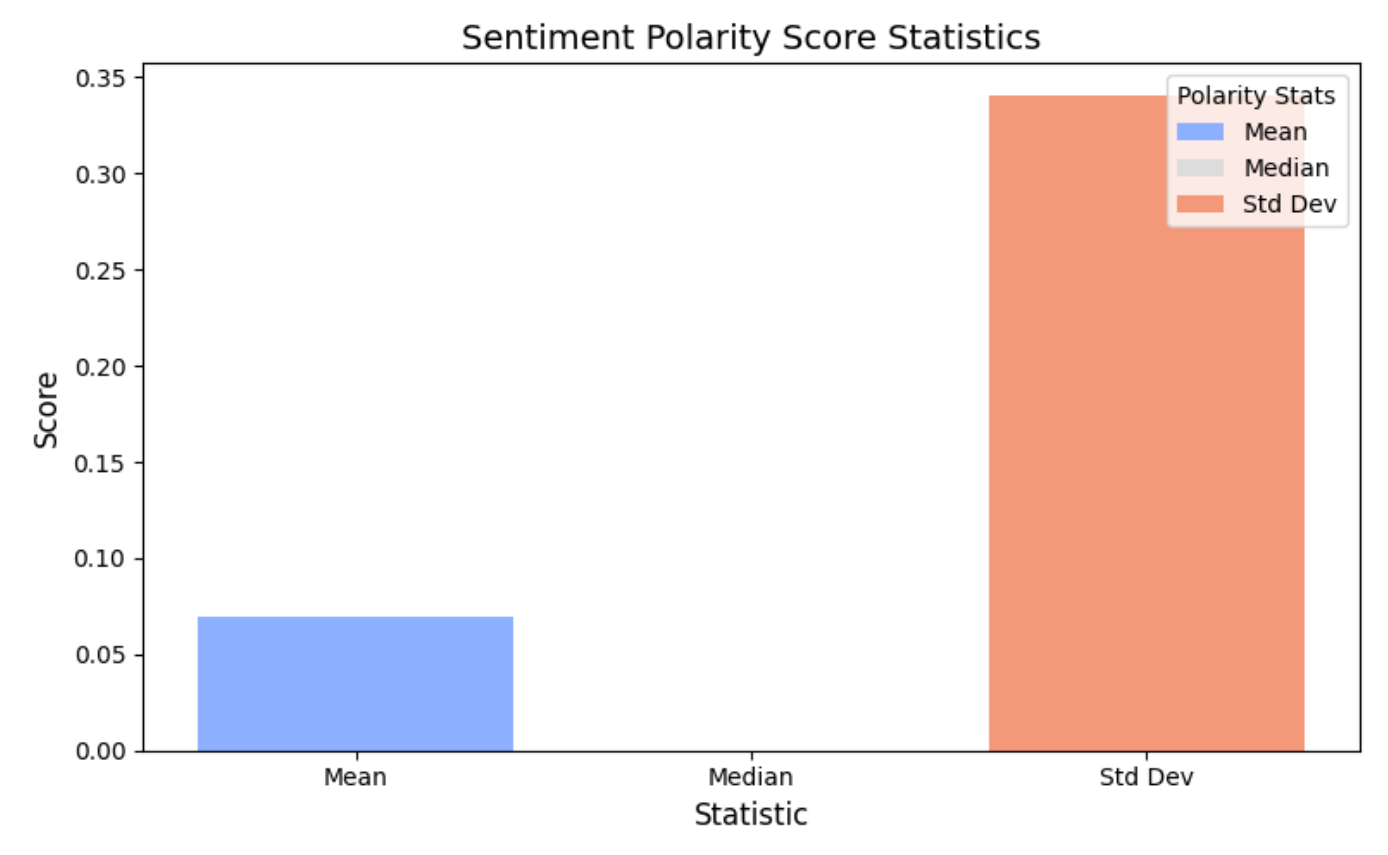
Engagement Distribution Num Comments



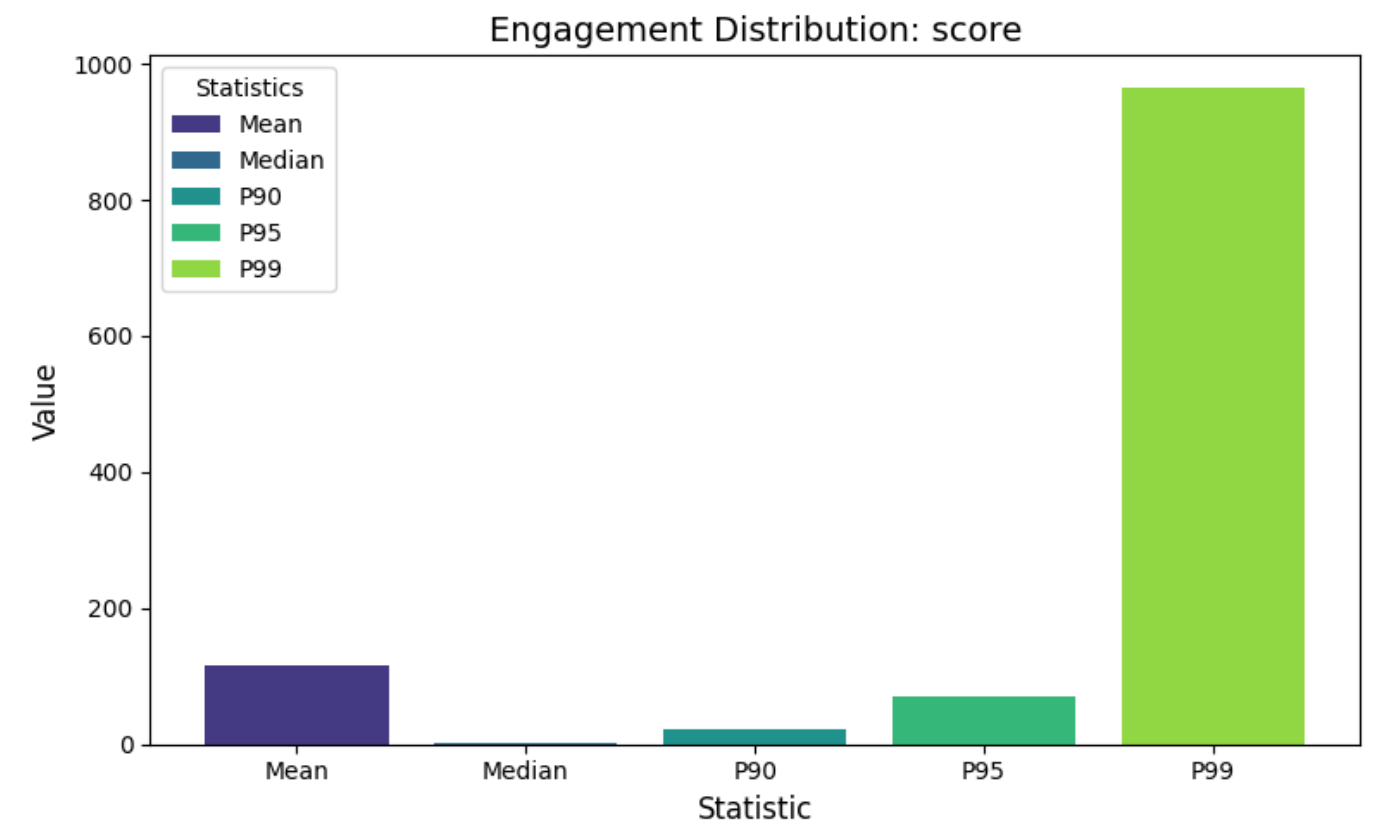
Sentiment Class Distribution



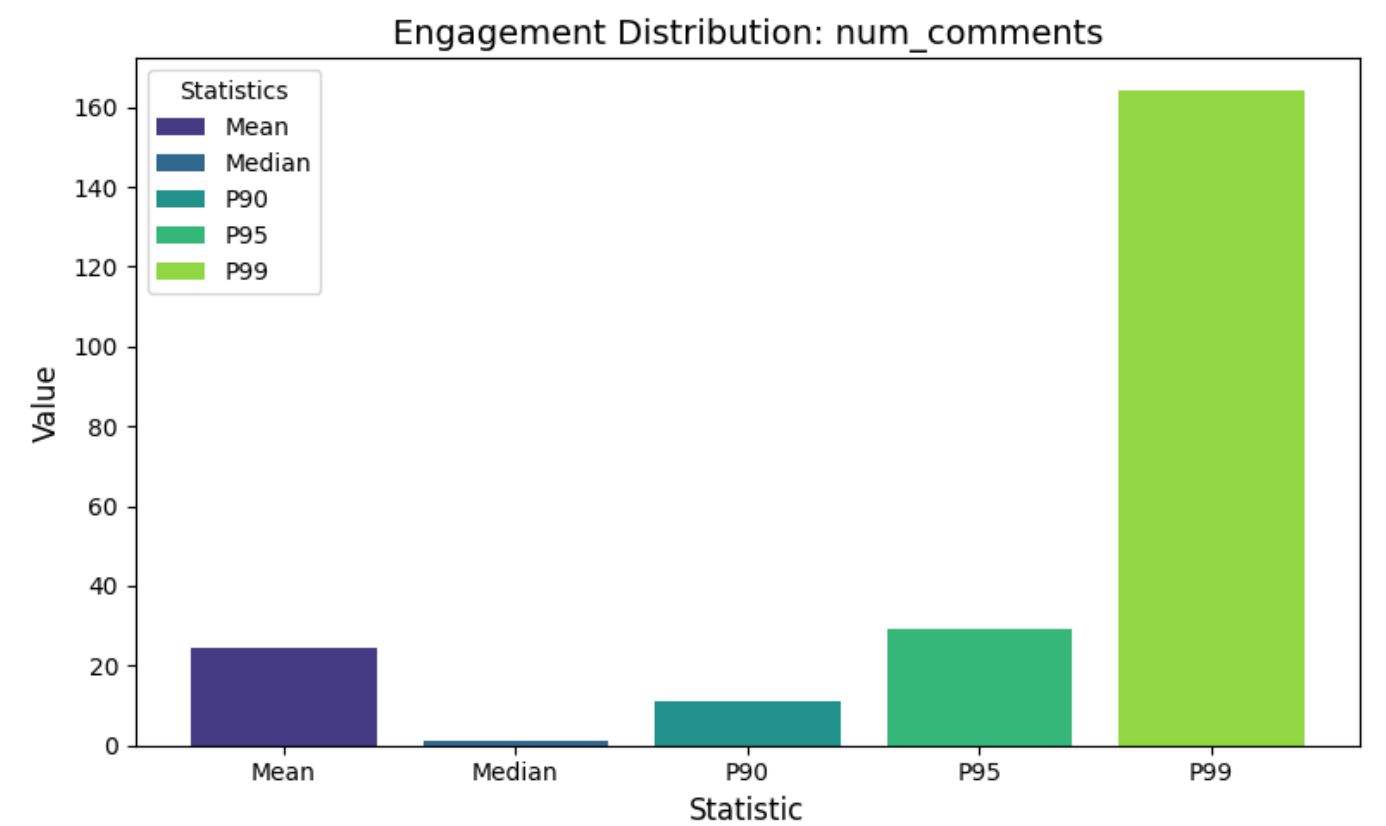
Sentiment Polarity Stats



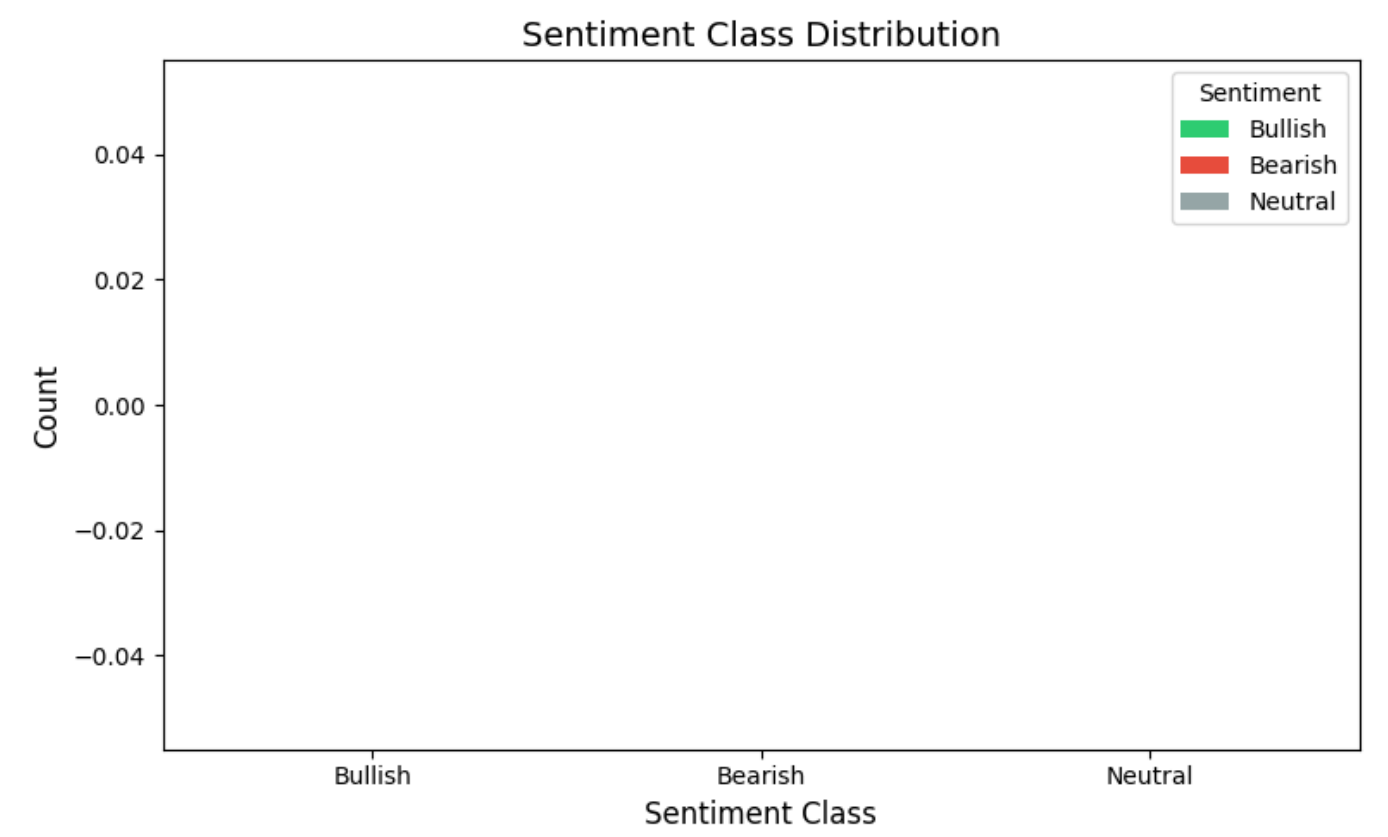
Engagement Distribution Score



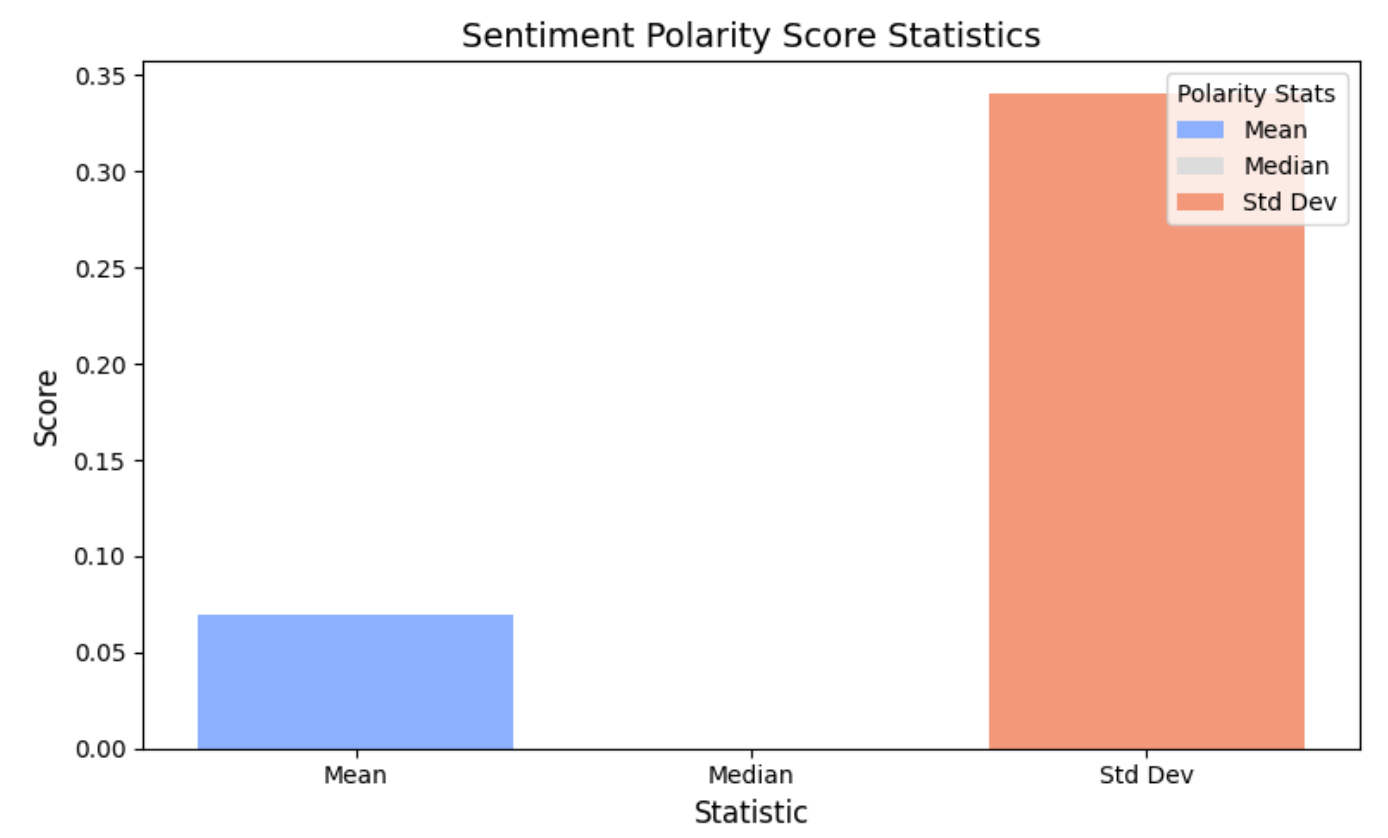
Engagement Distribution Num Comments



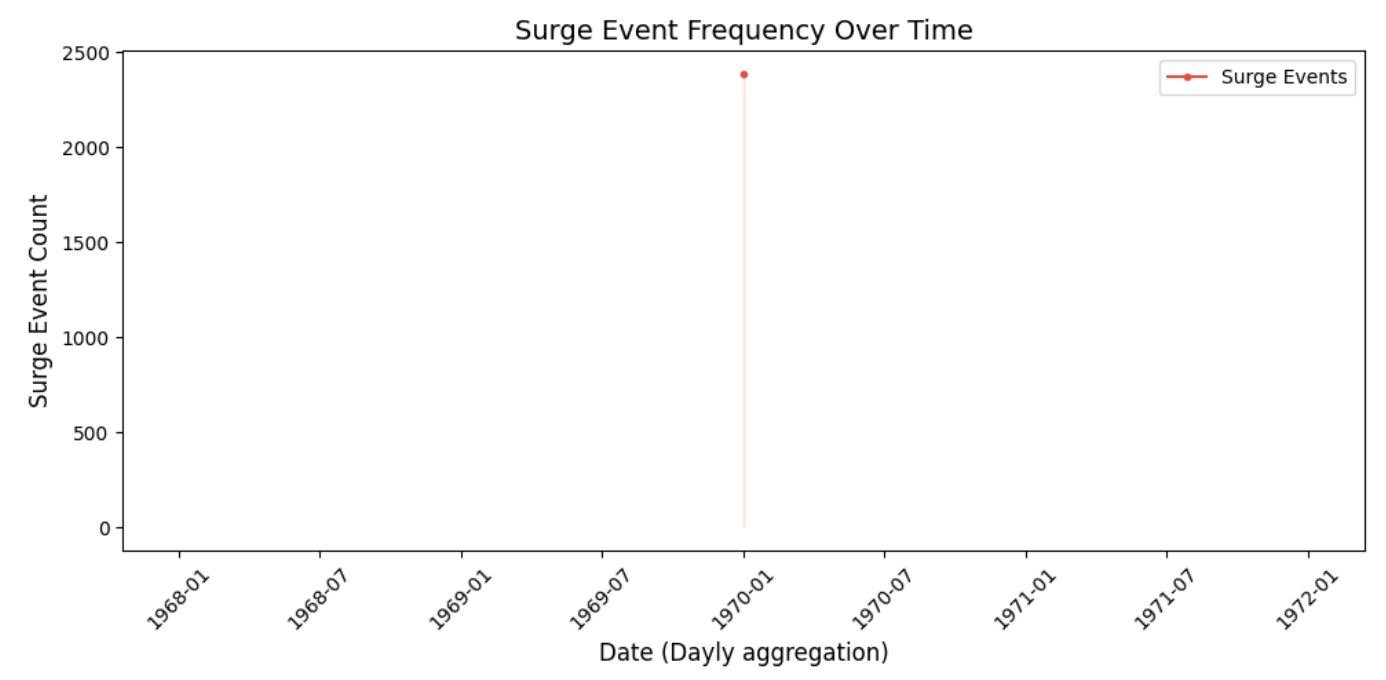
Sentiment Class Distribution



Sentiment Polarity Stats



Surge Frequency



Pipeline Performance

Stage Timings

Stage	Duration (s)
Dataset Discovery	42.15
API Feasibility	0.00
Dataset Preparation	144.30
Dataset Quality	504.49
Surge Analysis	263.02
Visualization	12.05
Total	966.02

Per-Dataset Processing Time

Dataset	Duration (s)
./data/StockMarket_subreddit.csv	29.36
./data/goyaladi_twitter-dataset/goyaladi_twitter-dataset.csv	8.68
./data/leukipp_reddit-finance-data/gme/submissions_reddit.csv	107.18
./data/leukipp_reddit-finance-data/investing/submissions_reddit.csv	15.16
./data/leukipp_reddit-finance-data/options/submissions_reddit.csv	10.27
./data/leukipp_reddit-finance-data/pennystocks/submissions_reddit.csv	18.95
./data/leukipp_reddit-finance-data/robinhood/submissions_reddit.csv	6.91

Dataset	Duration (s)
./data/leukipp_reddit-finance-data/robinhoodpennystocks/submissions_reddit.csv	7.99
./data/leukipp_reddit-finance-data/stockmarket/submissions_reddit.csv	14.84
./data/leukipp_reddit-finance-data/stocks/submissions_reddit.csv	23.26
./data/leukipp_reddit-finance-data/wallstreetbets/submissions_reddit.csv	261.88

Final Recommendation

Recommended Path: API Collection (reddit API)

Recommend API collection via 'reddit API' as the best data path. Key strengths: reasonable cost, supports surge label construction, historical data access available. API collection provides fresh, customizable data tailored to the prediction task. Surge analysis confirms viable definitions exist (48/90 configurations produce $\geq 2\%$ positive class).

Ranked Options

1. **reddit API** (overall score: 1.000)
- Low cost for data collection
 - Fast collection time
 - Supports surge label construction
 - Some fields require paid access: full historical archive, real-time streaming

Scoring Breakdown:

Criterion	Score	Weight	Comment
Cost	1.00	25%	Free access
Collection Time	1.00	20%	Very fast (≤ 1 hour)
Feasibility	1.00	25%	Supports surge label construction
Historical Access	1.00	15%	Historical data access available
Field Availability	1.00	15%	Rich field set (10+ fields)

2. **./data/leukipp_reddit-finance-data/pennystocks/submissions_reddit.csv** (overall score: 0.960)
- High data completeness with few missing values
 - Large dataset suitable for model training
 - Good temporal coverage with minimal gaps

Scoring Breakdown:

Criterion	Score	Weight	Comment
Completeness	1.00	25%	Excellent - very few missing values

Criterion	Score	Weight	Comment
Volume	0.80	20%	Good - 10k-100k records
Temporal Coverage	1.00	20%	No temporal gaps detected
Ticker Diversity	1.00	15%	Excellent - 50+ tickers covered
Risk	1.00	20%	No risks identified

3. `./data/leukipp_reddit-finance-data/gme/submissions_reddit.csv` (overall score: 0.960)

- High data completeness with few missing values
- Large dataset suitable for model training
- Good temporal coverage with minimal gaps
- 1 risk(s) identified: High missing data: columns ['selftext'] have >30% missing values, which may bias analysis or require imputation.

Scoring Breakdown:

Criterion	Score	Weight	Comment
Completeness	0.96	25%	Excellent - very few missing values
Volume	1.00	20%	Excellent - 100k+ records
Temporal Coverage	1.00	20%	No temporal gaps detected
Ticker Diversity	1.00	15%	Excellent - 50+ tickers covered
Risk	0.85	20%	Minor risks (1-2 issues)

4. `./data/leukipp_reddit-finance-data/wallstreetbets/submissions_reddit.csv` (overall score: 0.960)

- High data completeness with few missing values
- Large dataset suitable for model training
- Good temporal coverage with minimal gaps
- 1 risk(s) identified: High missing data: columns ['selftext'] have >30% missing values, which may bias analysis or require imputation.

Scoring Breakdown:

Criterion	Score	Weight	Comment
Completeness	0.96	25%	Excellent - very few missing values
Volume	1.00	20%	Excellent - 100k+ records
Temporal Coverage	1.00	20%	No temporal gaps detected
Ticker Diversity	1.00	15%	Excellent - 50+ tickers covered
Risk	0.85	20%	Minor risks (1-2 issues)

5. `./data/StockMarket_subreddit.csv` (overall score: 0.930)

- High data completeness with few missing values
- Large dataset suitable for model training
- Good temporal coverage with minimal gaps
- 1 risk(s) identified: Duplicate rows detected: 1103 duplicates (1.5%), which may inflate engagement statistics.

Scoring Breakdown:

Criterion	Score	Weight	Comment
Completeness	1.00	25%	Excellent - very few missing values
Volume	0.80	20%	Good - 10k-100k records
Temporal Coverage	1.00	20%	No temporal gaps detected
Ticker Diversity	1.00	15%	Excellent - 50+ tickers covered
Risk	0.85	20%	Minor risks (1-2 issues)

6. **./data/leukipp_reddit-finance-data/investing/submissions_reddit.csv** (overall score: 0.920)

- High data completeness with few missing values
- Large dataset suitable for model training
- Good temporal coverage with minimal gaps
- 1 risk(s) identified: High missing data: columns ['link_flair_text'] have >30% missing values, which may bias analysis or require imputation.

Scoring Breakdown:

Criterion	Score	Weight	Comment
Completeness	0.96	25%	Excellent - very few missing values
Volume	0.80	20%	Good - 10k-100k records
Temporal Coverage	1.00	20%	No temporal gaps detected
Ticker Diversity	1.00	15%	Excellent - 50+ tickers covered
Risk	0.85	20%	Minor risks (1-2 issues)

7. **./data/leukipp_reddit-finance-data/options/submissions_reddit.csv** (overall score: 0.920)

- High data completeness with few missing values
- Large dataset suitable for model training
- Good temporal coverage with minimal gaps
- 1 risk(s) identified: High missing data: columns ['link_flair_text'] have >30% missing values, which may bias analysis or require imputation.

Scoring Breakdown:

Criterion	Score	Weight	Comment
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Criterion	Score	Weight	Comment
Completeness	0.96	25%	Excellent - very few missing values
Volume	0.80	20%	Good - 10k-100k records
Temporal Coverage	1.00	20%	No temporal gaps detected
Ticker Diversity	1.00	15%	Excellent - 50+ tickers covered
Risk	0.85	20%	Minor risks (1-2 issues)

8. **./data/leukipp_reddit-finance-data/robinhood/submissions_reddit.csv** (overall score: 0.920)

- High data completeness with few missing values
- Large dataset suitable for model training
- Good temporal coverage with minimal gaps
- 1 risk(s) identified: High missing data: columns ['link_flair_text'] have >30% missing values, which may bias analysis or require imputation.

Scoring Breakdown:

Criterion	Score	Weight	Comment
Completeness	0.96	25%	Excellent - very few missing values
Volume	0.80	20%	Good - 10k-100k records
Temporal Coverage	1.00	20%	No temporal gaps detected
Ticker Diversity	1.00	15%	Excellent - 50+ tickers covered
Risk	0.85	20%	Minor risks (1-2 issues)

9. **./data/leukipp_reddit-finance-data/stockmarket/submissions_reddit.csv** (overall score: 0.920)

- High data completeness with few missing values
- Large dataset suitable for model training
- Good temporal coverage with minimal gaps
- 1 risk(s) identified: High missing data: columns ['selftext'] have >30% missing values, which may bias analysis or require imputation.

Scoring Breakdown:

Criterion	Score	Weight	Comment
Completeness	0.96	25%	Excellent - very few missing values
Volume	0.80	20%	Good - 10k-100k records
Temporal Coverage	1.00	20%	No temporal gaps detected
Ticker Diversity	1.00	15%	Excellent - 50+ tickers covered
Risk	0.85	20%	Minor risks (1-2 issues)

10. **./data/leukipp_reddit-finance-data/stocks/submissions_reddit.csv** (overall score: 0.920)

- High data completeness with few missing values
- Large dataset suitable for model training
- Good temporal coverage with minimal gaps
- 1 risk(s) identified: High missing data: columns ['link_flair_text'] have >30% missing values, which may bias analysis or require imputation.

Scoring Breakdown:

Criterion	Score	Weight	Comment
Completeness	0.96	25%	Excellent - very few missing values
Volume	0.80	20%	Good - 10k-100k records
Temporal Coverage	1.00	20%	No temporal gaps detected
Ticker Diversity	1.00	15%	Excellent - 50+ tickers covered
Risk	0.85	20%	Minor risks (1-2 issues)

11. **./data/leukipp_reddit-finance-data/robinhoodpennystocks/submissions_reddit.csv** (overall score: 0.910)

- High data completeness with few missing values
- Large dataset suitable for model training
- Good temporal coverage with minimal gaps
- 1 risk(s) identified: High missing data: columns ['link_flair_text', 'selftext'] have >30% missing values, which may bias analysis or require imputation.

Scoring Breakdown:

Criterion	Score	Weight	Comment
Completeness	0.92	25%	Excellent - very few missing values
Volume	0.80	20%	Good - 10k-100k records
Temporal Coverage	1.00	20%	No temporal gaps detected
Ticker Diversity	1.00	15%	Excellent - 50+ tickers covered
Risk	0.85	20%	Minor risks (1-2 issues)

12. **./data/goyaladi_twitter-dataset/goyaladi_twitter-dataset.csv** (overall score: 0.810)

- High data completeness with few missing values
- Large dataset suitable for model training
- Good temporal coverage with minimal gaps
- 1 risk(s) identified: No ticker column 'ticker' found and ticker extraction from text yielded no results: per-ticker engagement normalization cannot be performed.

Scoring Breakdown:

Criterion	Score	Weight	Comment
Completeness	1.00	25%	Excellent - very few missing values
Volume	0.80	20%	Good - 10k-100k records
Temporal Coverage	1.00	20%	No temporal gaps detected
Ticker Diversity	0.20	15%	Very limited - fewer than 3 tickers
Risk	0.85	20%	Minor risks (1-2 issues)

13. **twitter API** (overall score: 0.770)

- Fast collection time
- Supports surge label construction
- No historical data access - requires prospective collection
- Some fields require paid access: impression_count, full archive search, quote_count

Scoring Breakdown:

Criterion	Score	Weight	Comment
Cost	0.50	25%	Moderate cost (\$50-\$200)
Collection Time	1.00	20%	Very fast (≤ 1 hour)
Feasibility	1.00	25%	Supports surge label construction
Historical Access	0.30	15%	No historical access - prospective only
Field Availability	1.00	15%	Rich field set (10+ fields)